PickleVerse Software Design Description

Report 4 · Current architecture data design and proposed completion work

Working draft for team and reviewer discussion. Source snapshot c32f2a4c0a54 on 23 October 2026. No game/backend execution or code edits were performed. PKV is a working abbreviation; official code, names and approvals remain unconfirmed.

1 Design boundary and architecture

The current design combines a Unity rendering/gameplay client, an ASP.NET Core data-authority backend, PostgreSQL, Redis, Photon Fusion Shared, SMTP and S3. The backend is authoritative for commands and durable account/economy state. Real-time simulation still belongs to a peer StateAuthority. “Server” method naming inside a peer NetworkBehaviour does not establish a dedicated trusted server.

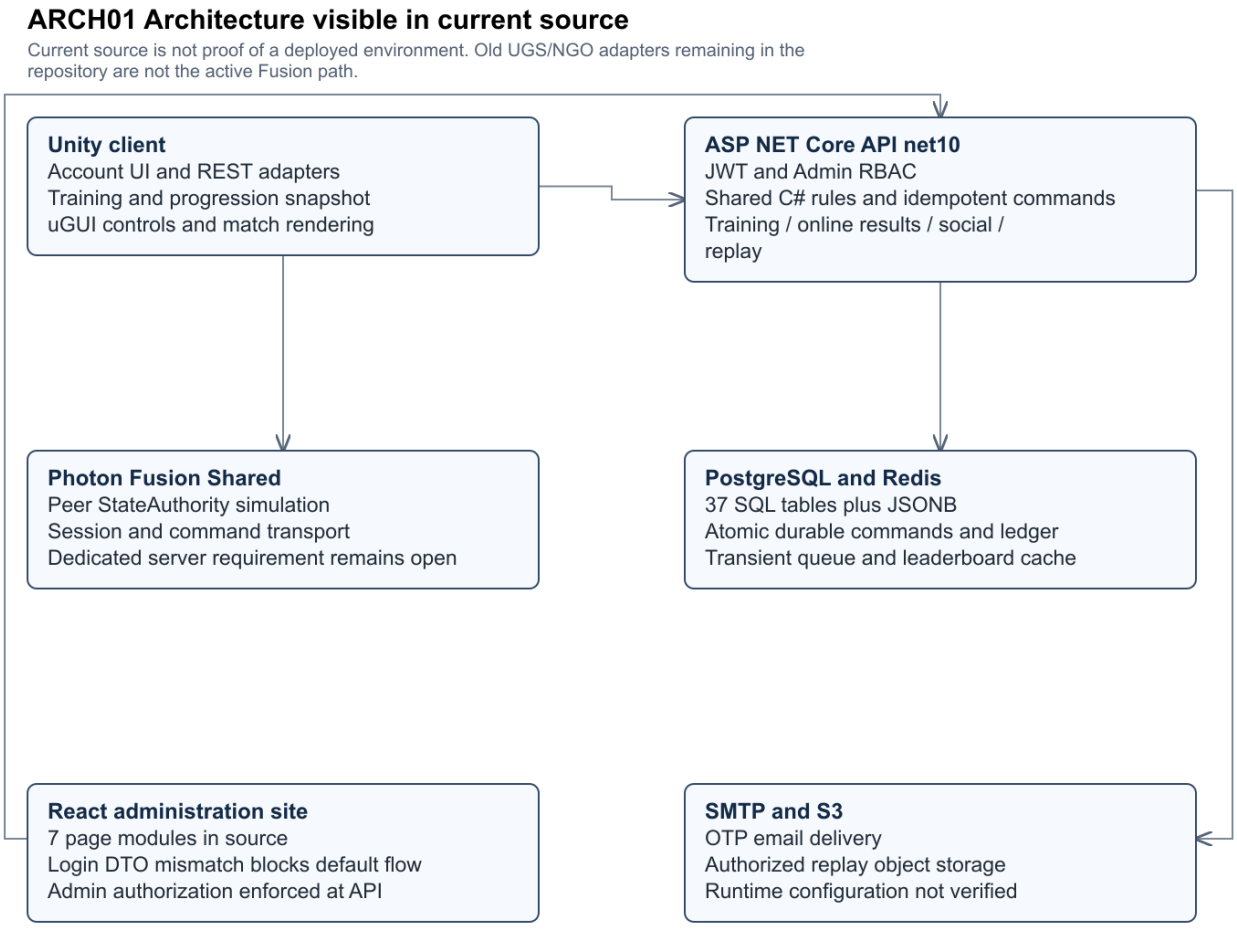


Figure ARCH01 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

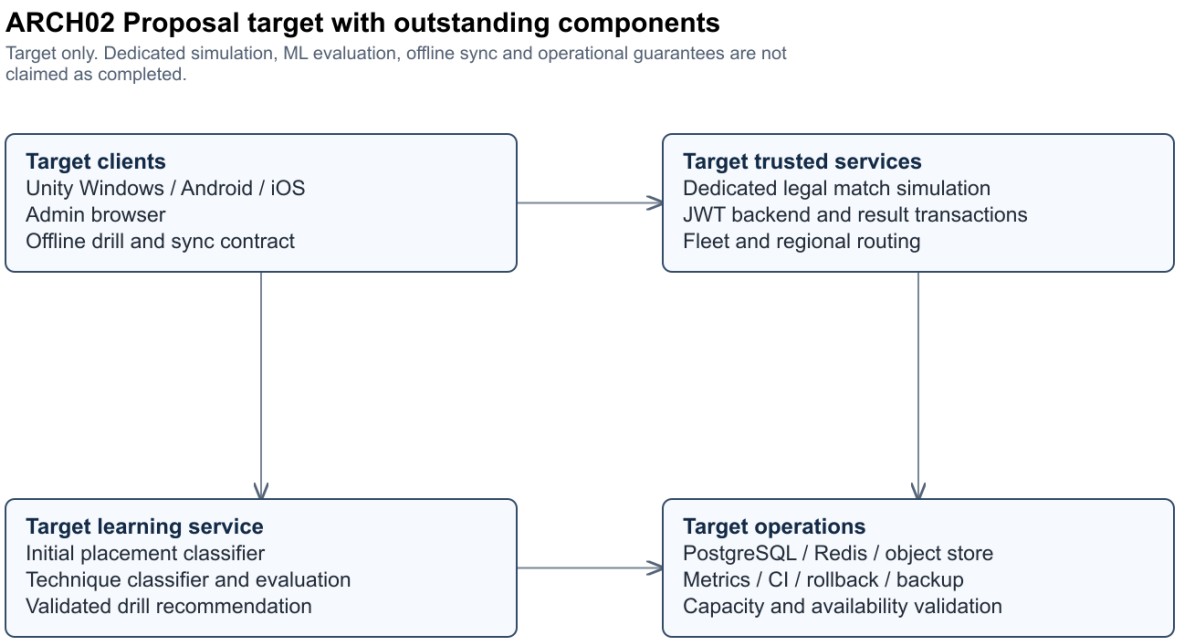


Figure ARCH02 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Component	Version or source contract	Responsibility and limit
Unity client	6000.6.2f1; URP 17.6.0; Input System 1.20.0; uGUI 2.6.0	Windows/mobile rendering, controls, local simulation and REST adapters; runtime performance not measured.
Real-time network	Photon Fusion Shared in PhotonSession.cs	Shared peer authority and session transport; exact installed Fusion package version requires package/vendor manifest confirmation.
Backend	net10.0; ASP.NET Core; EF models plus raw SQL	JWT, validated commands, catalog, training, results, social and replay.
Backend libraries	StackExchange.Redis 3.4.5; MailKit 4.17.0; AWSSDK.S3 4.0.104.5	Source package references, not proof of deployed service configuration.
Data	PostgreSQL source DDL; Redis runtime service	37 source tables; 27 EF classes plus raw-SQL additions. Deployed engine/version/migration state must be recorded.
Admin browser	React 18; Vite 5; TypeScript 5; Tailwind 3; React Query 5	Users/CMS/matches/moderation/analytics source exists; default auth DTO mismatch blocks integration.

2 Components and trust boundaries

Account and API transport

AccountRuntime creates the API client before scene load. UnityWebRequest supplies JSON and Bearer headers with timeout/cancellation. AccountService rotates refresh tokens on 401, keeps tokens in memory and invalidates stale account-generation responses. Password recovery is not established by the observed registration/login UI.

Catalog and progression

GameCatalogService reads configured lesson/mission/cosmetic data and rule settings. ProgressionRuntime is an in-memory view of server state. GameProgress stores aggregate JSONB and revision; existing LocalPlayerStore classes are not evidence that the active runtime is offline-capable.

Command transaction

OnlineGameService locks account/profile state, validates command ID and payload fingerprint, applies shared C# rules, persists profile/wallet/objective/ledger state, inserts command identity and commits. Identical retries return current state; conflicting reuse fails. PostgreSQL transactions, not UI state, protect exactly-once awards.

Training validation

The server creates attempt IDs, checks ownership/catalog/prerequisites, validates monotonic finite measurements and plausible elapsed time, and determines completion/review reward. It does not observe every rendered contact; measurement honesty remains a security boundary.

Ranked queue and rooms

Redis maintains search/reservation and ready policy; canonical roster and match data persist in PostgreSQL. Search mode/region/protocol/membership must agree. Configured ranked-bot fallback is a product-policy decision to disclose, not a human matchmaking guarantee.

Online results

GameMatches stores canonical session/roster/phase. GameMatchPlayers stores acceptance, report fingerprints and heartbeat. All required humans must accept; result reports must agree and satisfy score/roster rules. Finalization updates every participant transactionally. A forged-but-agreed result still needs the dedicated authority solution or an approved security tradeoff.

Social ranking and replay

Online social APIs provide real paged accounts/relationships and Elo standings. LeaderboardCache derives versioned sorted sets from PostgreSQL. Replay services authorize participants and validate bounded gzip chunks; metadata and S3 keys/hashes live in SQL. Friend-to-room invitations are distinct from friendship requests and code-based joining.

Admin and analytics

AdminController enforces the Admin JWT role. The React auth payload/response contract, enum/SQL status mismatch, configured-catalog publication and metric definitions remain open. UI pages cannot be used as evidence that these flows pass end-to-end.

3 Design decisions and alternatives

ID	Decision	Alternatives	Choice and rationale	Owner	Trace
D1	Common account UUID and REST snapshot	Keep isolated local/UGS identity or integrate backend UUID	Use account UUID to scope all state and reject stale-session responses; source implemented.	M1	F01/F04 ARCH01
D2	Durable command ledger	Client rewards or idempotent server transaction	Use PostgreSQL transaction plus payload fingerprint and ledger; shared pure C# rules avoid divergent reward logic.	M4	F03/F10 DATA01
D3	Real-time authority	Dedicated headless authority or Photon Shared peer	Source uses Shared; proposal dedicated authority remains unresolved. Seek explicit approval or implement trusted simulation.	M3	F09 SR01 ARCH02

ID	Decision	Alternatives	Choice and rationale	Owner	Trace
D4	Runtime content configuration	Hard-coded drills or configured catalog	Use coded validated config and version hash; admin publish path still needs completion.	M1	F03/F16 SR06
D5	Relational plus JSONB state	Fully normalized replacement or hybrid schema	Document actual 37-table hybrid design; keep JSON structures explicit and Redis ephemeral. Do not deploy the separate 71-table candidate implicitly.	M4	F04/F10 ERD00

Owner slots and technology deviations are proposals for team confirmation, not inferred personal contributions or reviewer approval.

4 Relational and logical data design

The authoritative schema for this audit is src/Database.sql at the stated HEAD. Every table and column is represented in ERD01–ERD11 and Data_Dictionary_current.csv. Each diagram includes complete attributes for its primary tables and reference stubs for other pages. Dashed references are cross-page SQL relations, not new tables. The Mermaid source carries parent/child optionality; the relationship register below lists declared FK columns. Unique/composite constraints must be read with the original DDL.

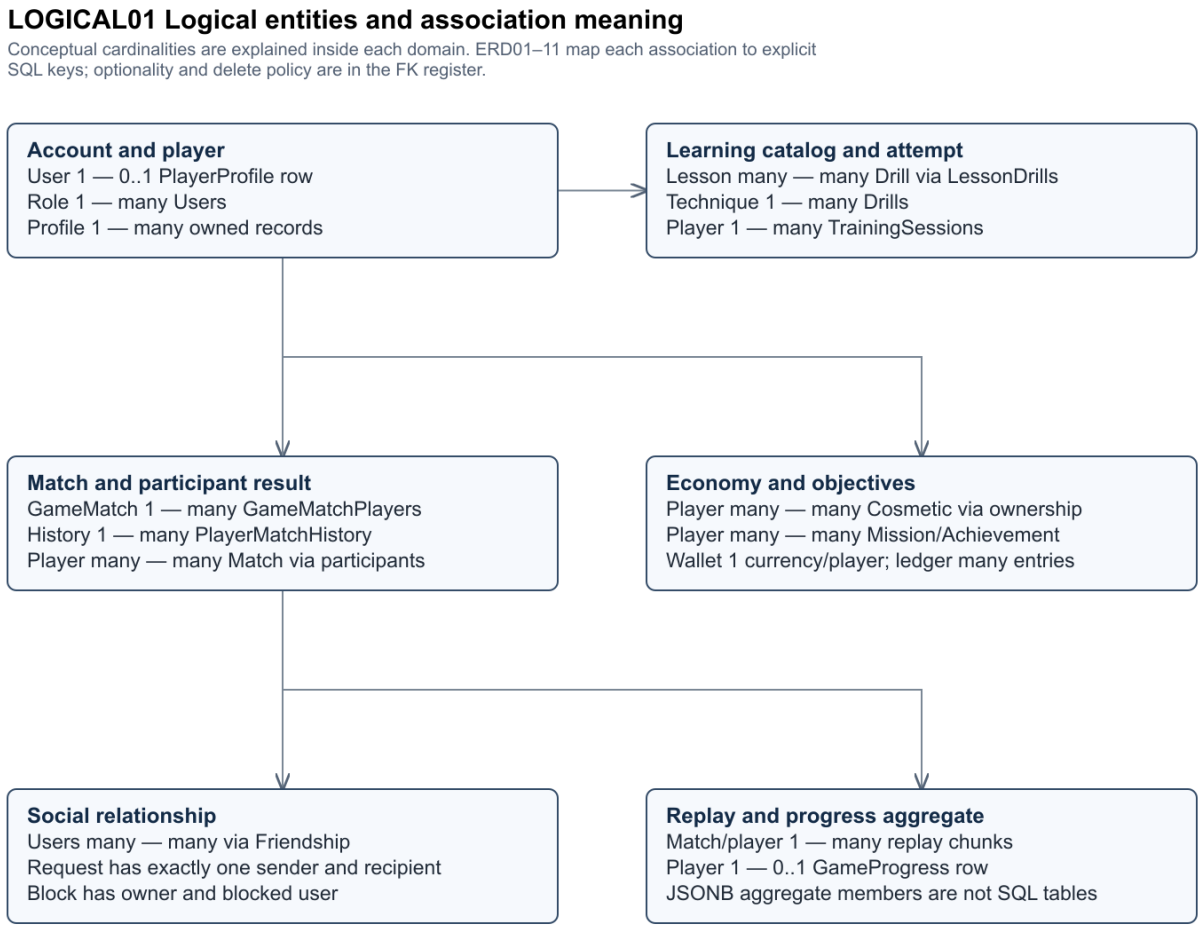


Figure LOGICAL01 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

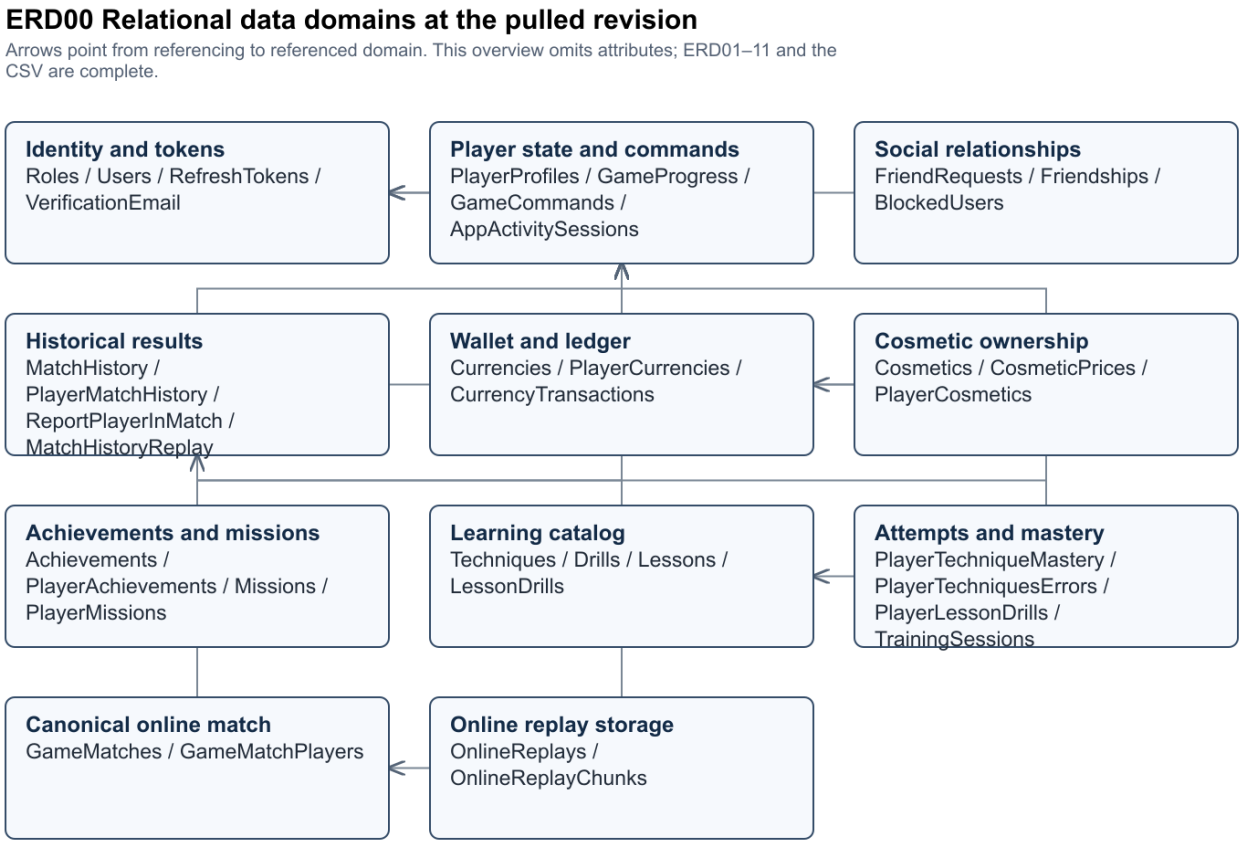


Figure ERD00 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Current schema domain 1

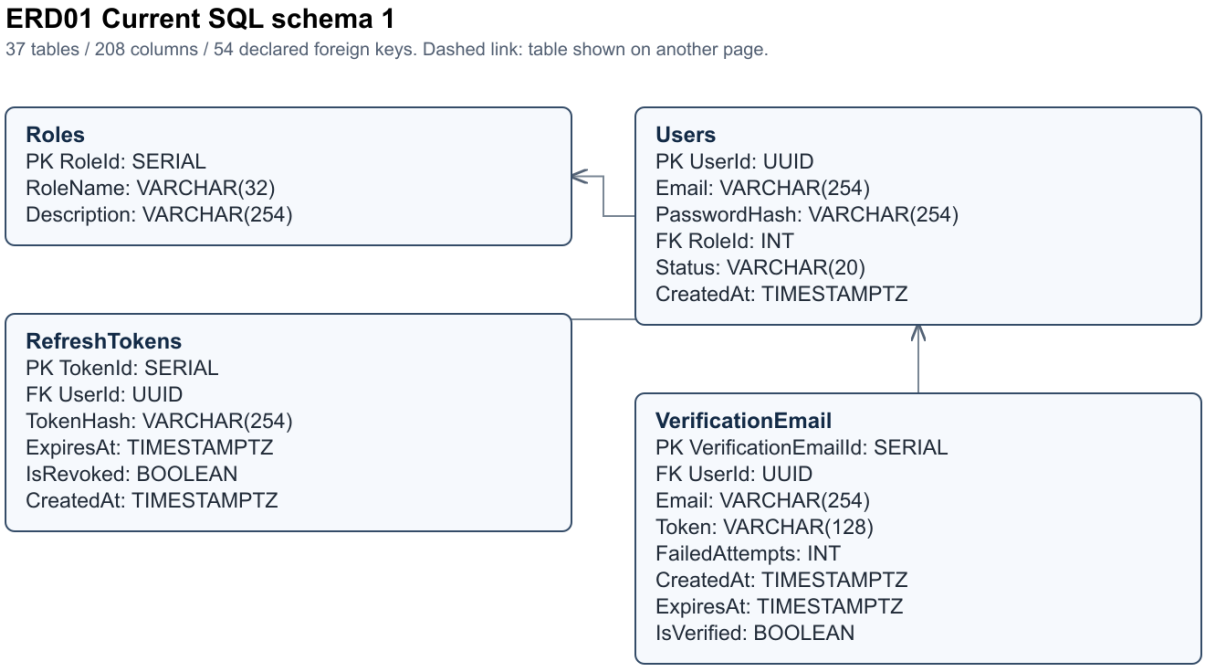


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Current schema domain 2

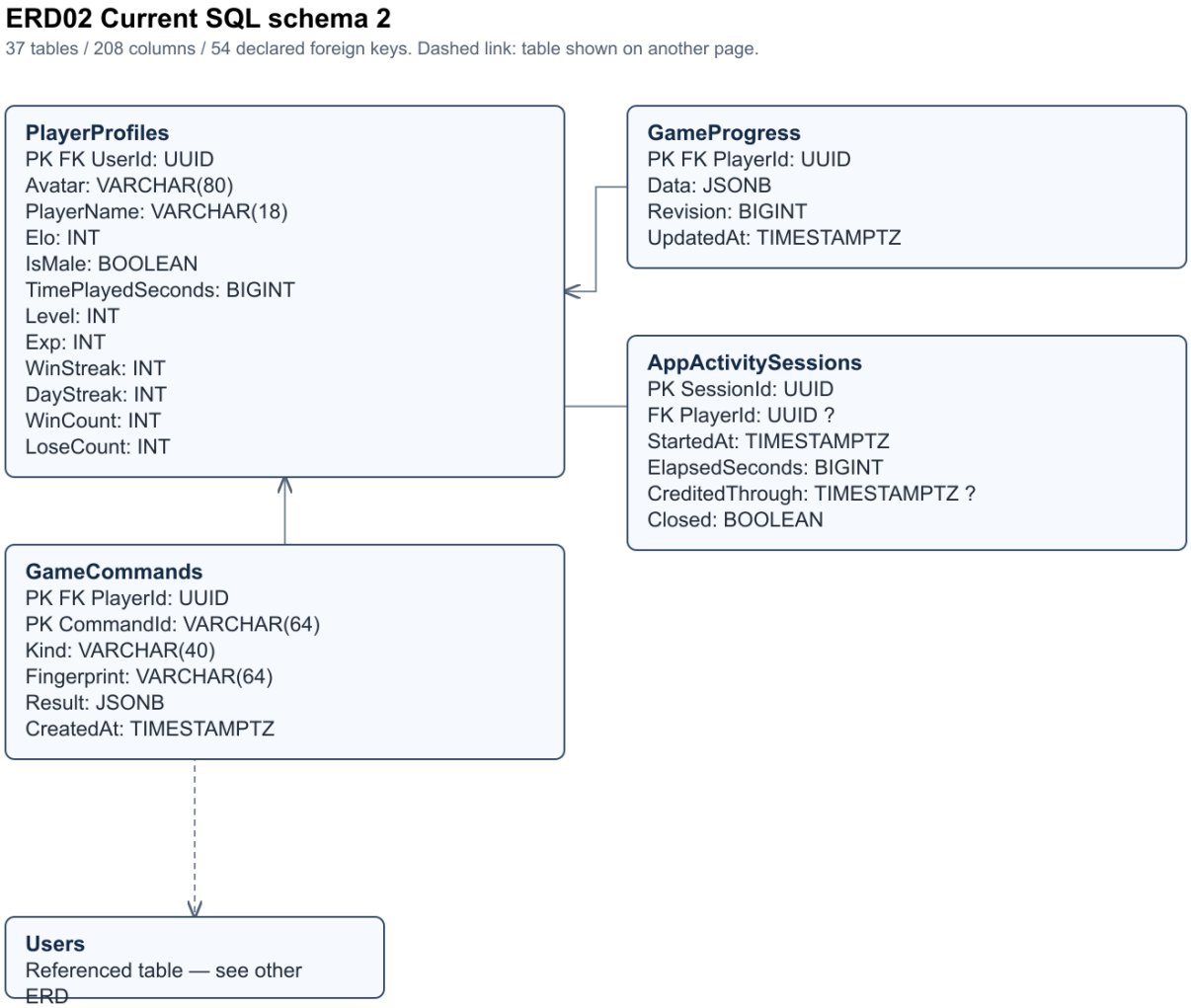


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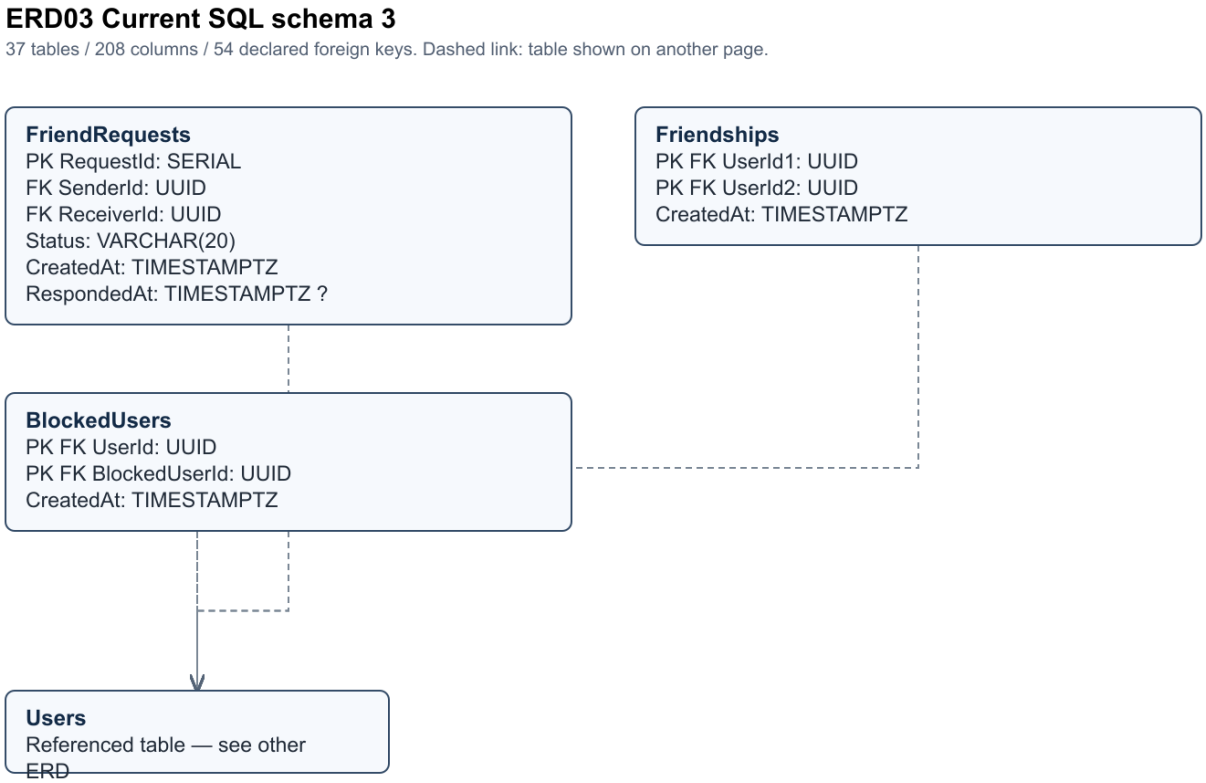


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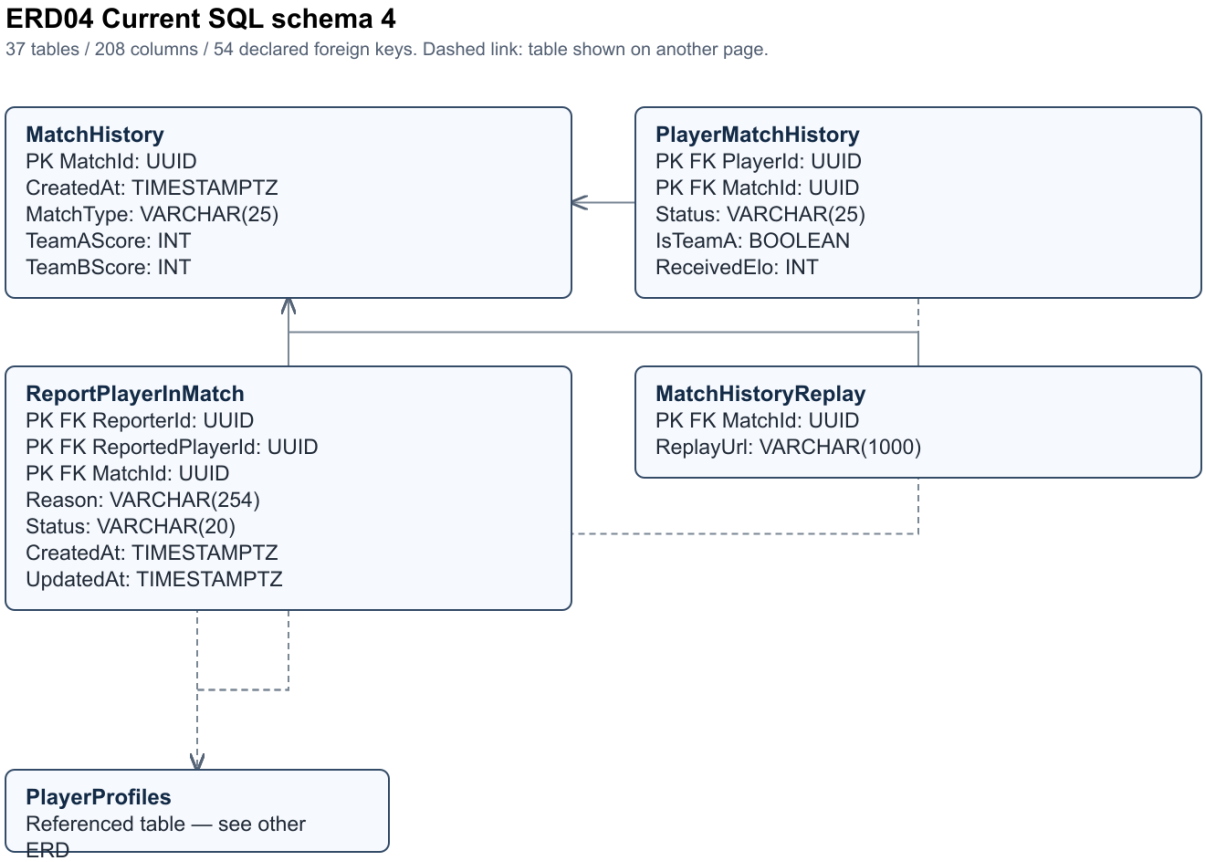


Figure ERD04 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

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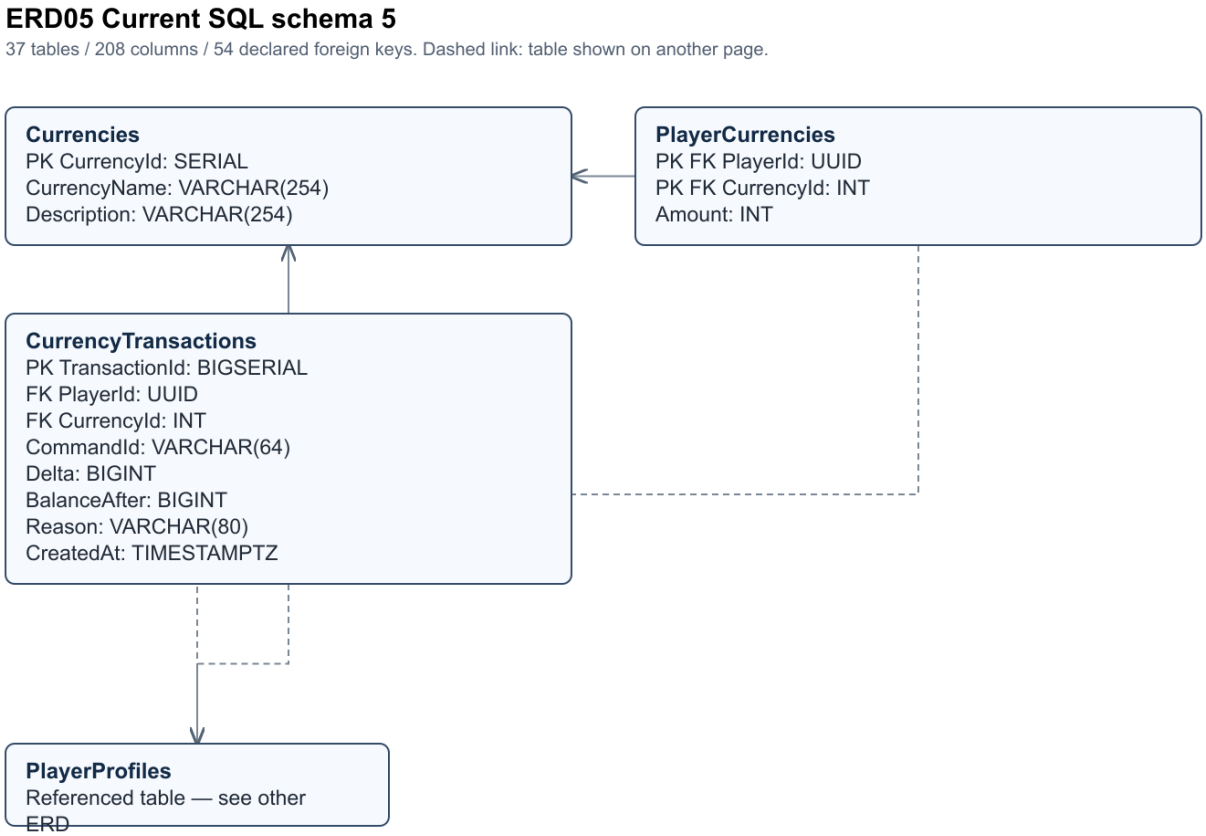


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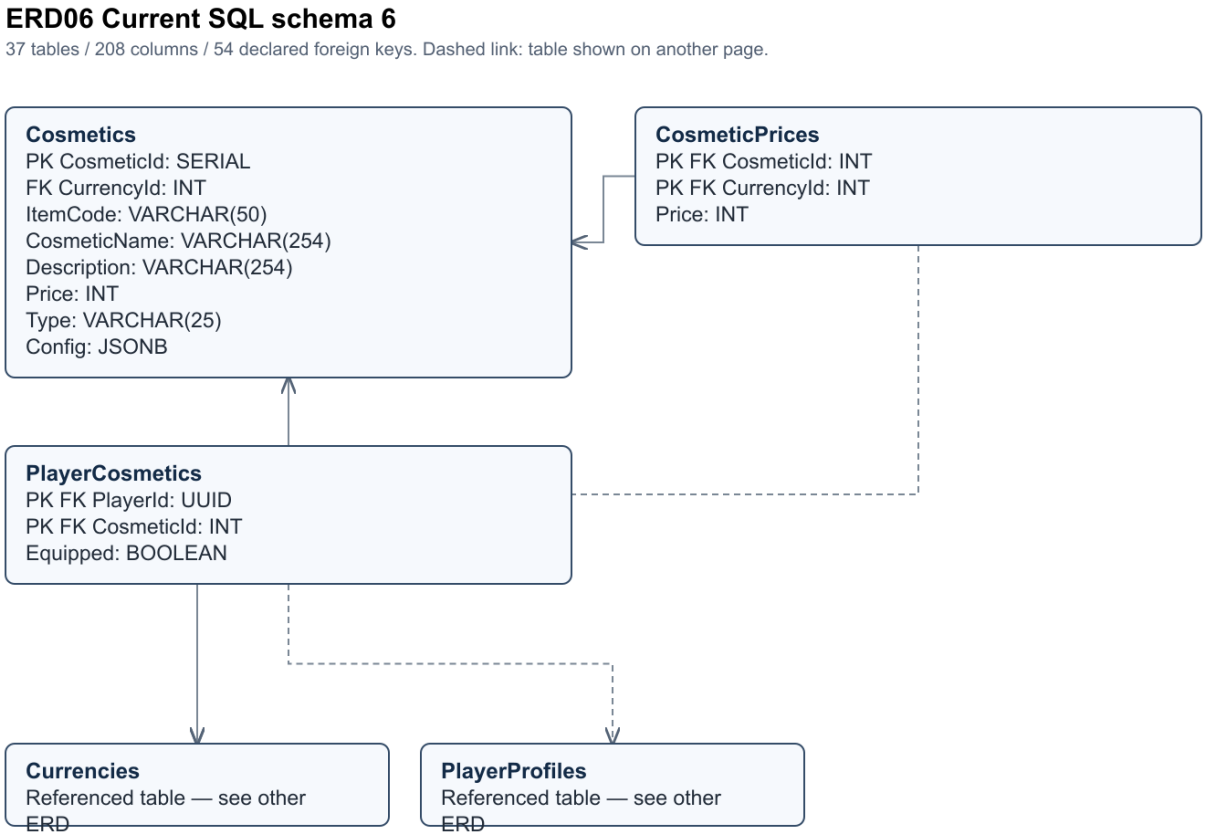


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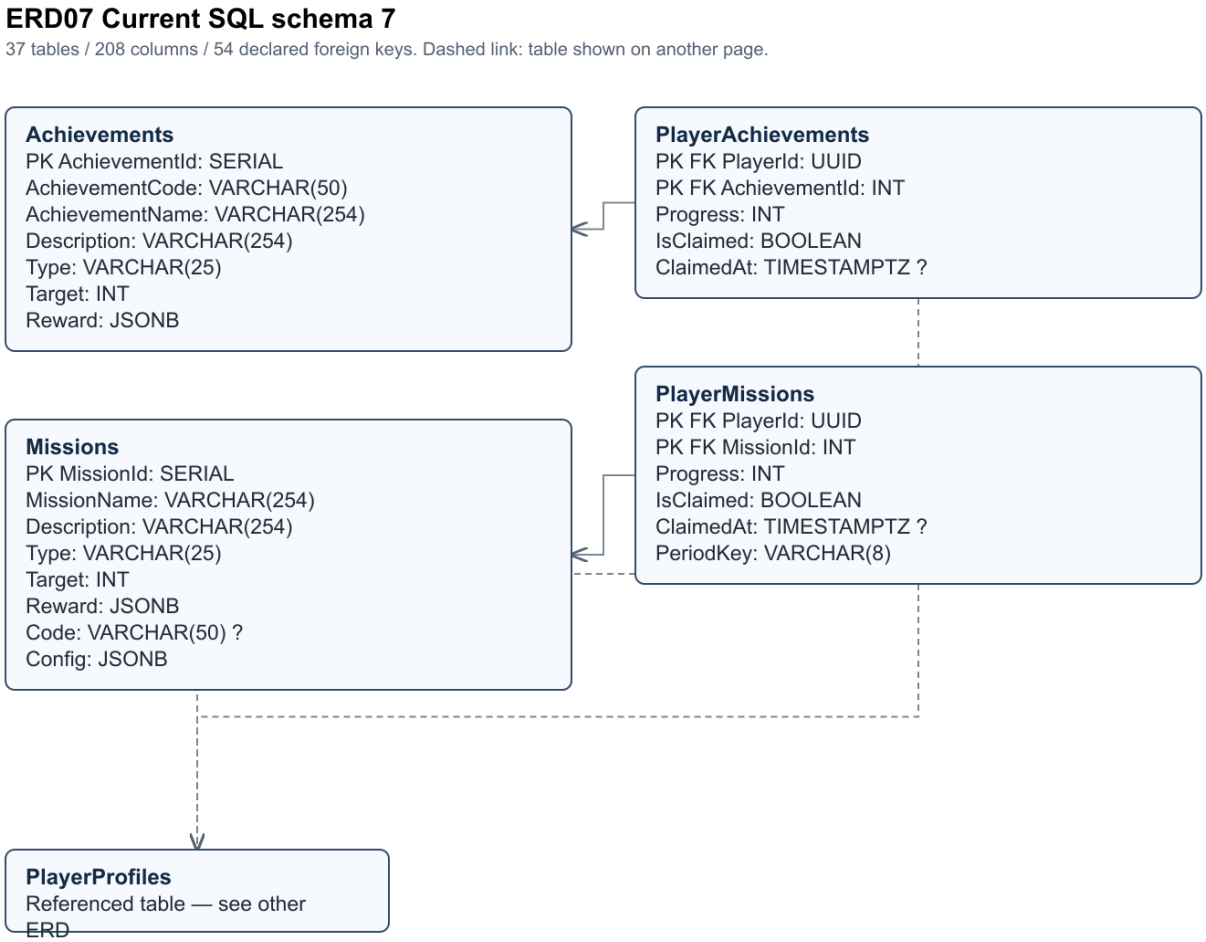


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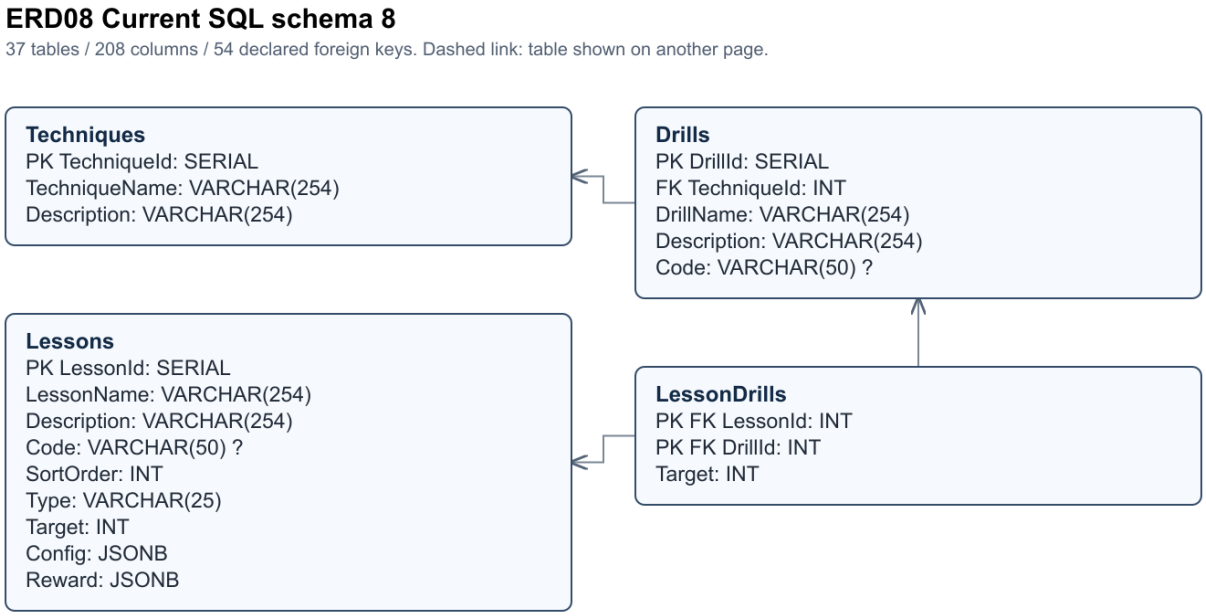


Figure ERD08 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Current schema domain 9

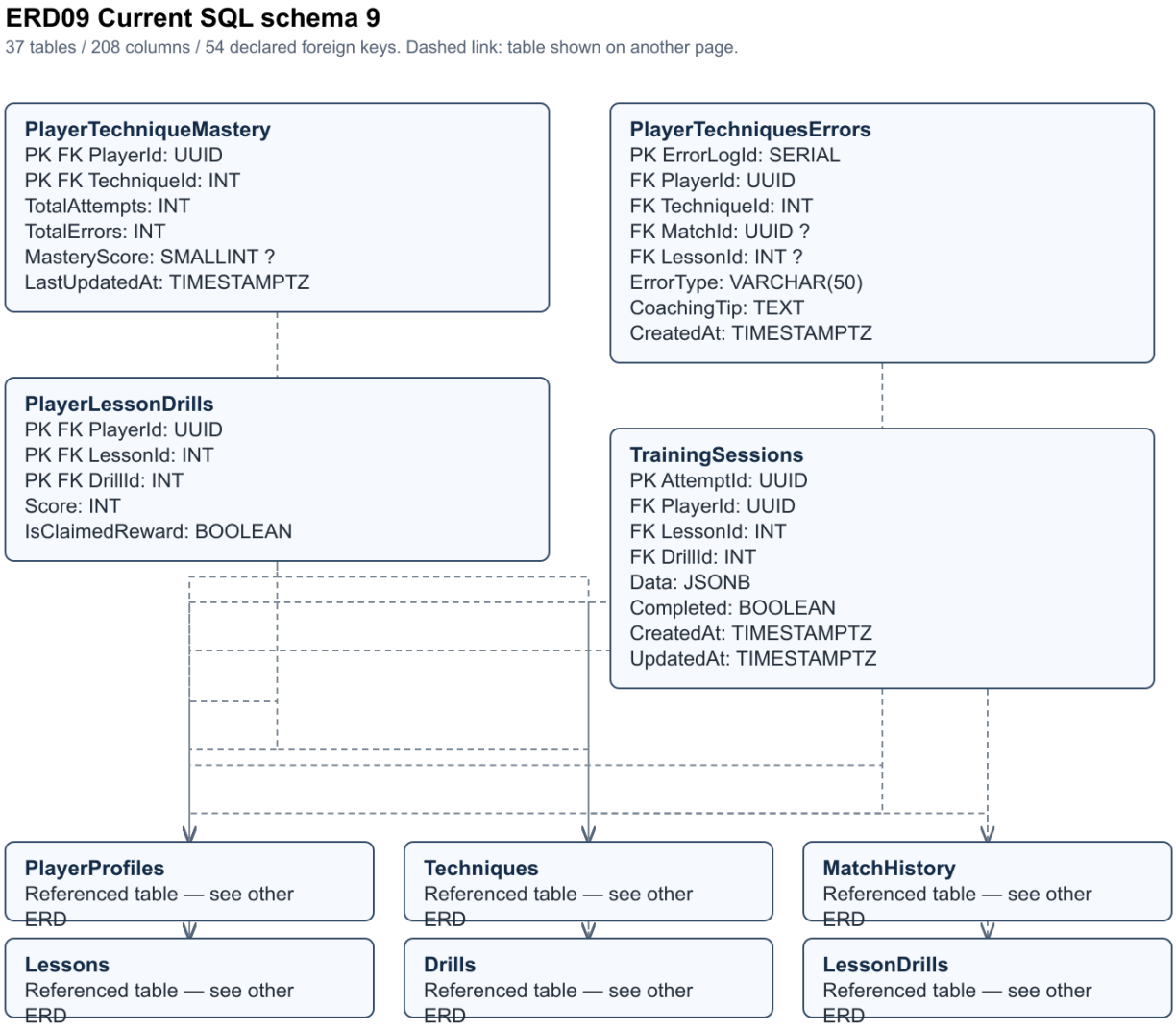


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Current schema domain 10

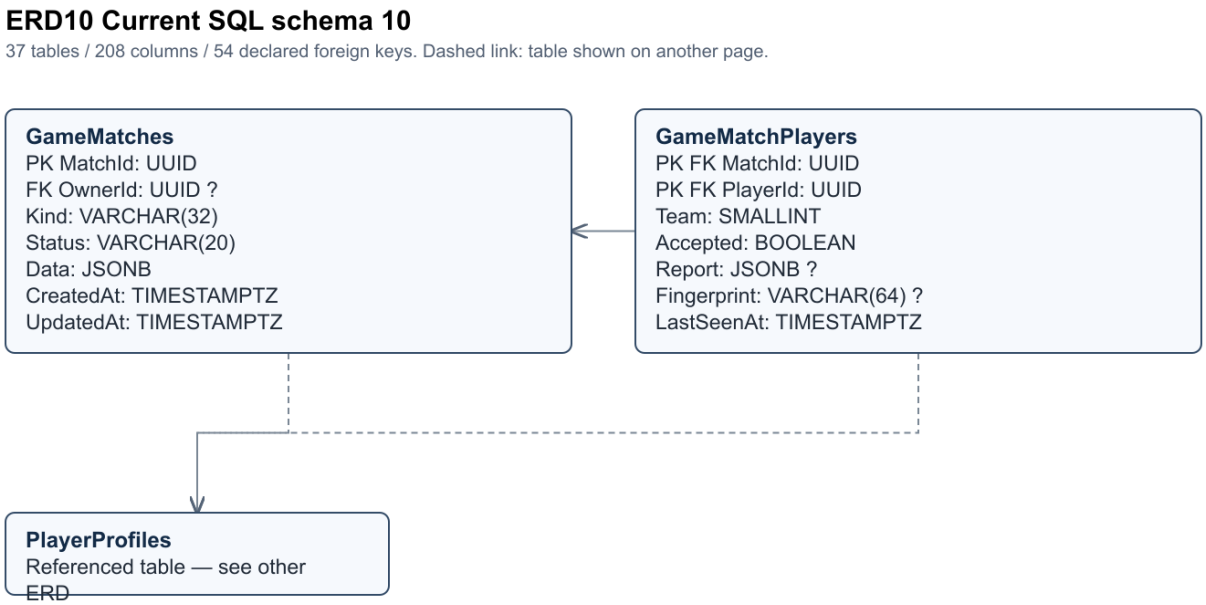


Figure ERD10 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Current schema domain 11

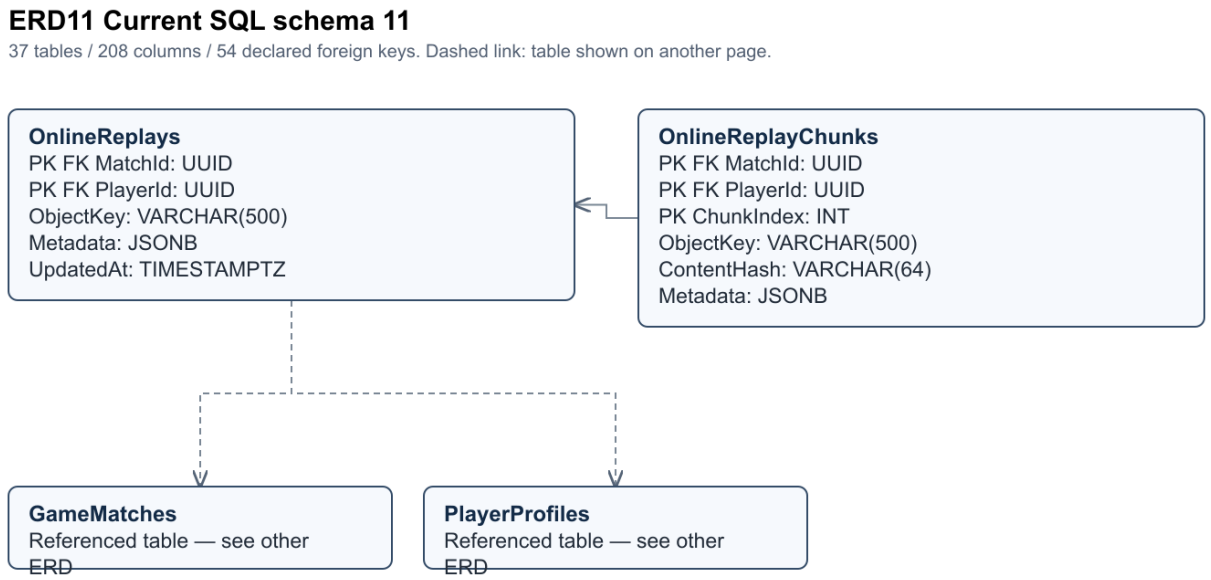


Figure ERD11 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Foreign key relationship register

Referencing columns	Referenced columns	Parent per child	Delete
Users.RoleId	Roles.RoleId	1	NO ACTION
RefreshTokens.UserId	Users.UserId	1	CASCADE
VerificationEmail.UserId	Users.UserId	1	CASCADE
PlayerProfiles.UserId	Users.UserId	1	CASCADE
FriendRequests.SenderId	Users.UserId	1	CASCADE
FriendRequests.ReceiverId	Users.UserId	1	CASCADE
Friendships.UserId1	Users.UserId	1	CASCADE
Friendships.UserId2	Users.UserId	1	CASCADE
BlockedUsers.UserId	Users.UserId	1	CASCADE
BlockedUsers.BlockedUserId	Users.UserId	1	CASCADE
PlayerMatchHistory.PlayerId	PlayerProfiles.UserId	1	NO ACTION
PlayerMatchHistory.MatchId	MatchHistory.MatchId	1	CASCADE
ReportPlayerInMatch.ReporterId	PlayerProfiles.UserId	1	CASCADE
ReportPlayerInMatch.ReportedPlayerId	PlayerProfiles.UserId	1	CASCADE
ReportPlayerInMatch.MatchId	MatchHistory.MatchId	1	CASCADE
MatchHistoryReplay.MatchId	MatchHistory.MatchId	1	CASCADE
PlayerCurrencies.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerCurrencies.CurrencyId	Currencies.CurrencyId	1	NO ACTION
Cosmetics.CurrencyId	Currencies.CurrencyId	1	NO ACTION
PlayerCosmetics.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerCosmetics.CosmeticId	Cosmetics.CosmeticId	1	NO ACTION
PlayerAchievements.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerAchievements.AchievementId	Achievements.AchievementId	1	NO ACTION
PlayerMissions.PlayerId	PlayerProfiles.UserId	1	CASCADE

Referencing columns	Referenced columns	Parent per child	Delete
PlayerMissions.MissionId	Missions.MissionId	1	NO ACTION
Drills.TechniqueId	Techniques.TechniqueId	1	CASCADE
LessonDrills.LessonId	Lessons.LessonId	1	CASCADE
LessonDrills.DrillId	Drills.DrillId	1	CASCADE
PlayerTechniqueMastery.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerTechniqueMastery.TechniqueId	Techniques.TechniqueId	1	CASCADE
PlayerTechniquesErrors.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerTechniquesErrors.TechniqueId	Techniques.TechniqueId	1	CASCADE
PlayerTechniquesErrors.MatchId	MatchHistory.MatchId	0..1	SET NULL
PlayerTechniquesErrors.LessonId	Lessons.LessonId	0..1	SET NULL
PlayerLessonDrills.PlayerId	PlayerProfiles.UserId	1	CASCADE
PlayerLessonDrills.LessonId	Lessons.LessonId	1	CASCADE
PlayerLessonDrills.DrillId	Drills.DrillId	1	CASCADE
CosmeticPrices.CosmeticId	Cosmetics.CosmeticId	1	CASCADE
CosmeticPrices.CurrencyId	Currencies.CurrencyId	1	NO ACTION
GameProgress.PlayerId	PlayerProfiles.UserId	1	CASCADE
GameCommands.PlayerId	PlayerProfiles.UserId	1	CASCADE
TrainingSessions.PlayerId	PlayerProfiles.UserId	1	CASCADE
TrainingSessions.LessonId	Lessons.LessonId	1	NO ACTION
TrainingSessions.DrillId	Drills.DrillId	1	NO ACTION
TrainingSessions.LessonId,DrillId	LessonDrills.LessonId,DrillId	1	NO ACTION
AppActivitySessions.PlayerId	PlayerProfiles.UserId	0..1	CASCADE
GameMatches.OwnerId	PlayerProfiles.UserId	0..1	SET NULL
GameMatchPlayers.MatchId	GameMatches.MatchId	1	CASCADE
GameMatchPlayers.PlayerId	PlayerProfiles.UserId	1	CASCADE
CurrencyTransactions.PlayerId	PlayerProfiles.UserId	1	CASCADE
CurrencyTransactions.CurrencyId	Currencies.CurrencyId	1	NO ACTION
OnlineReplays.MatchId	GameMatches.MatchId	1	CASCADE
OnlineReplays.PlayerId	PlayerProfiles.UserId	1	CASCADE
OnlineReplayChunks.MatchId,PlayerId	OnlineReplays.MatchId,PlayerId	1	CASCADE

JSONB transient state and object payloads

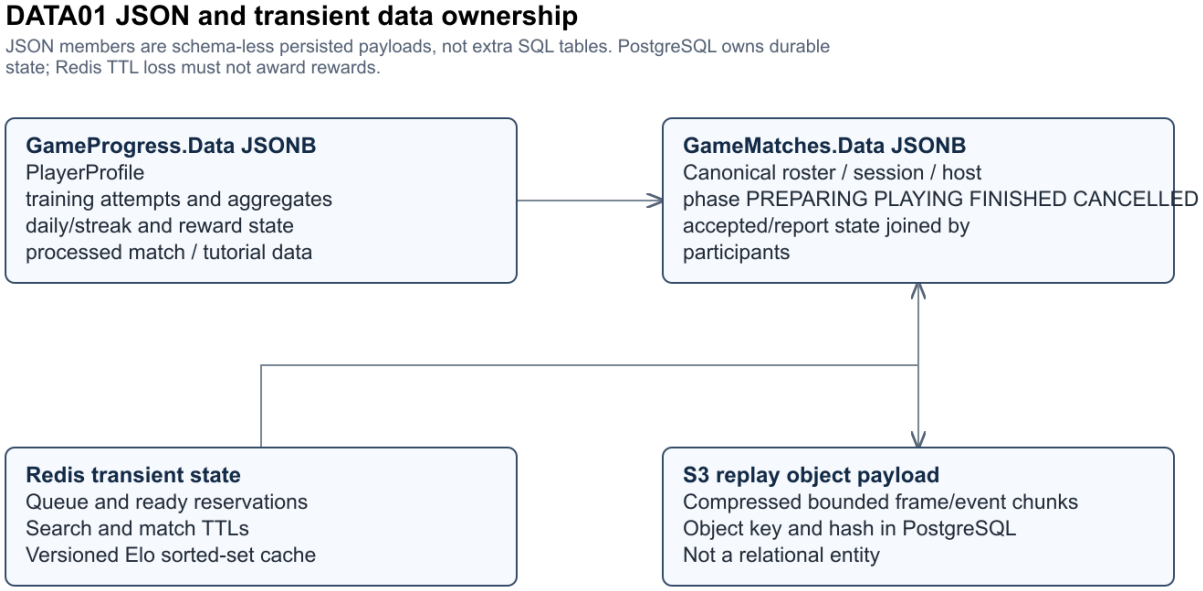


Figure DATA01 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

GameProgress.Data, GameMatches.Data and catalog Config columns carry embedded aggregate/configuration documents. They do not create implicit relational foreign keys. Attempt counters, errors and reward state are represented in shared serializable models; revisions/command IDs provide concurrency/idempotency boundaries. Redis queue TTLs and leaderboard epochs are operational data. S3 bytes are external objects referenced by SQL keys/hashes.

The 71-table src/NOREADDATABSE.sql is an alternative candidate, retained in Data_Dictionary_candidate.csv and historical diagram sources for comparison. It is not merged with the current ERD and is not labelled deployed. EF has 27 model classes; the ten new raw-SQL tables are expected in the documented hybrid design.

5 Main entity lifecycles and guard transitions

The models below include accepted and rejected transitions for testing. Except the explicit LocalMatch enum, labels may be derived business-state abstractions, not literal database values. Target publication/appeal states are deliberately labelled incomplete. For online matches, data.phase PREPARING/PLAYING differs from the broader SQL Status STARTED; they must not be conflated. Terminal states exit through explicit new entities/replay/retry only as declared.

SM01 Account lifecycle

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F01 / WF01. Evidence: My project/Assets/Scripts/Services/AccountService.cs:44

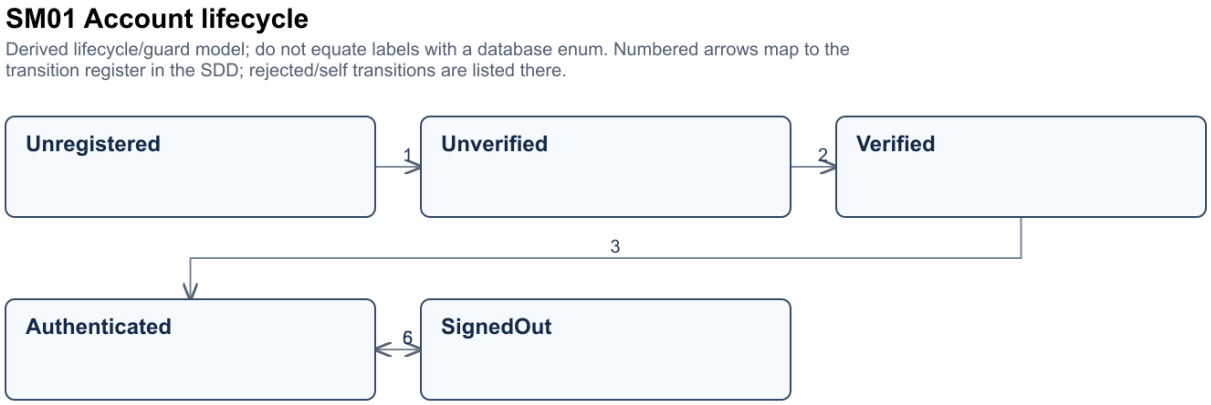


Figure SM01 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Unregistered	valid registration	Unverified	TC-SM01-01
2	Unverified	valid unexpired OTP	Verified	TC-SM01-02
3	Verified	valid login	Authenticated	TC-SM01-03
4	Authenticated	rotate valid refresh token	Authenticated	TC-SM01-04
5	Authenticated	logout/revoke	SignedOut	TC-SM01-05
6	SignedOut	valid login	Authenticated	TC-SM01-06
7	Unverified	reject wrong or expired OTP	Unverified	TC-SM01-07

SM02 Local match phase

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F06 / WF02. Evidence: My project/Assets/Scripts/LocalMatch/BotMatch.cs:11

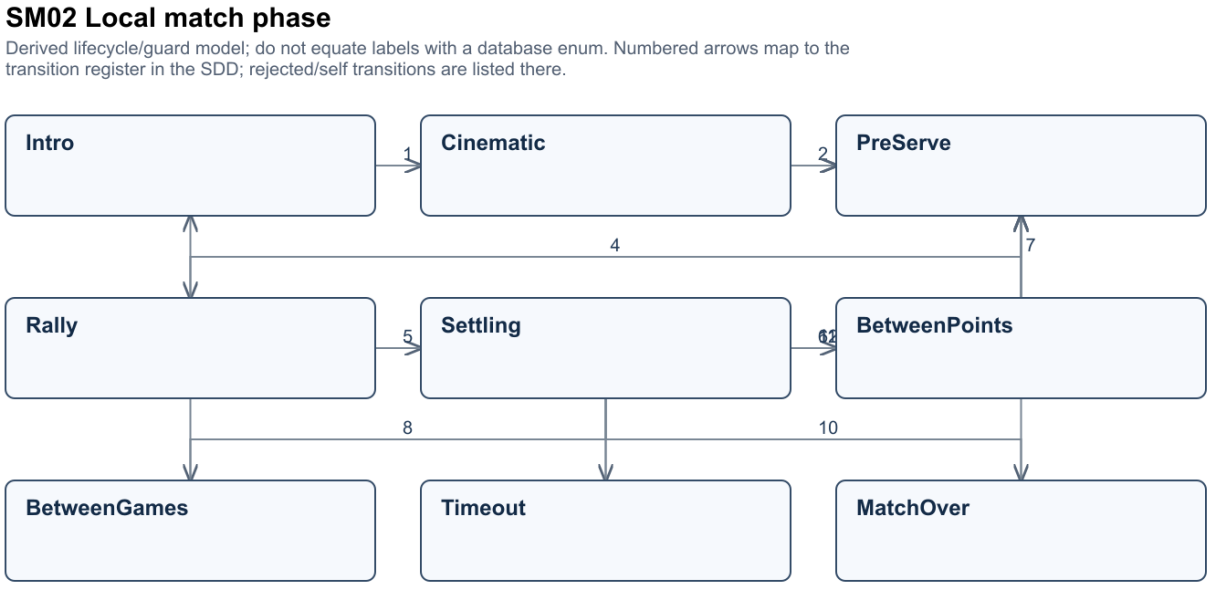


Figure SM02 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Intro	opening starts	Cinematic	TC-SM02-01
2	Cinematic	opening completes	PreServe	TC-SM02-02
3	Intro	opening skipped/unavailable	PreServe	TC-SM02-03
4	PreServe	legal serve	Rally	TC-SM02-04
5	Rally	fault or point	Settling	TC-SM02-05
6	Settling	point settled	BetweenPoints	TC-SM02-06
7	BetweenPoints	next point	PreServe	TC-SM02-07
8	Settling	game completed	BetweenGames	TC-SM02-08
9	BetweenGames	next game	PreServe	TC-SM02-09
10	Settling	match completed	MatchOver	TC-SM02-10
11	PreServe	legal timeout	Timeout	TC-SM02-11
12	Timeout	resume	PreServe	TC-SM02-12
13	MatchOver	explicit replay/start new match only	Intro	TC-SM02-13
14	PreServe	reject illegal serve	PreServe	TC-SM02-14

SM03 Training attempt

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F03 / WF01. Evidence: src/Backend/Services/Game/OnlineGameService.Training.cs:1

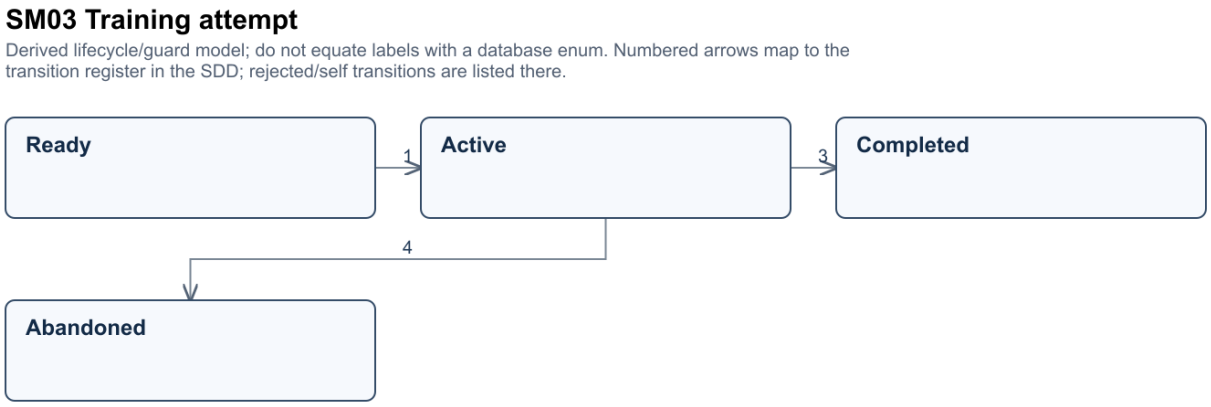


Figure SM03 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Ready	server attempt start	Active	TC-SM03-01
2	Active	valid checkpoint	Active	TC-SM03-02
3	Active	full valid trials submitted	Completed	TC-SM03-03
4	Active	leave/replaced attempt	Abandoned	TC-SM03-04
5	Active	reject foreign/non-monotonic trial	Active	TC-SM03-05
6	Completed	idempotent completion/review has no first reward	Completed	TC-SM03-06

SM04 Tutorial progress

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F02 / WF01. Evidence: My project/Assets/Scripts/LocalMatch/TutorialDirector.cs:11

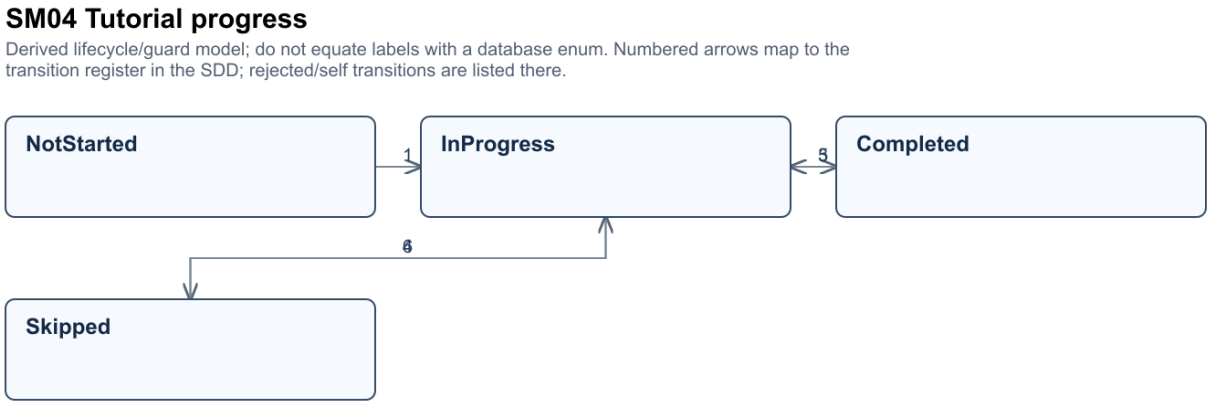


Figure SM04 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	NotStarted	open tutorial	InProgress	TC-SM04-01
2	InProgress	step satisfied/replay step	InProgress	TC-SM04-02
3	InProgress	all steps completed	Completed	TC-SM04-03
4	InProgress	confirm skip	Skipped	TC-SM04-04
5	Completed	explicit replay	InProgress	TC-SM04-05
6	Skipped	explicit replay	InProgress	TC-SM04-06

SM05 Skill queue and acceptance

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F07 / WF03. Evidence: src/Backend/Services/Game/OnlineGameService.OnlineMatches.cs:182

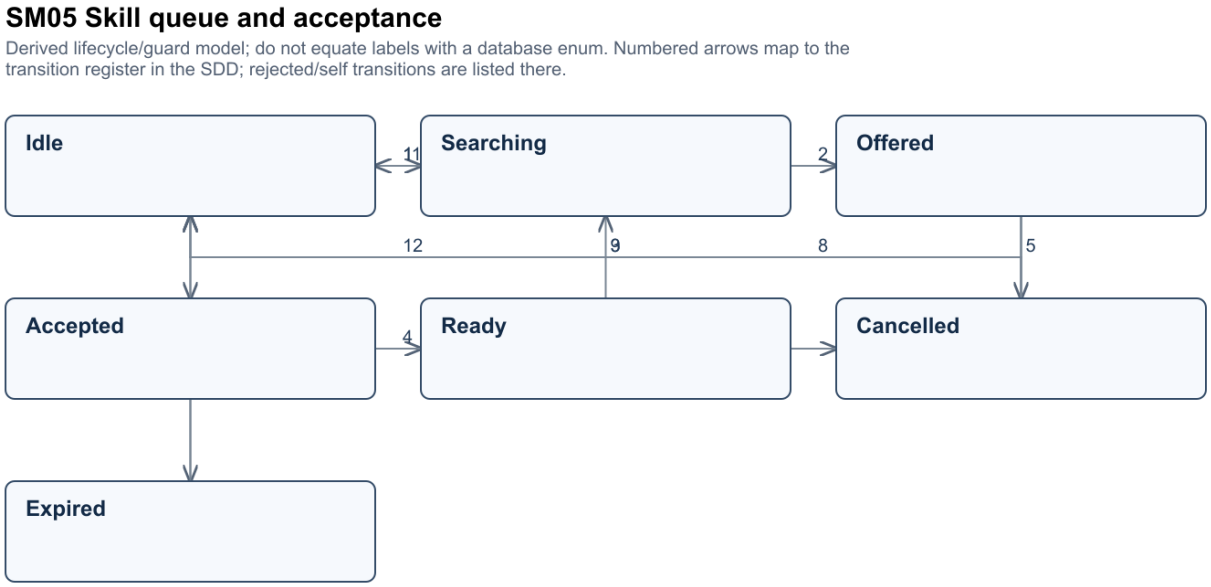


Figure SM05 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test

#	From	Trigger and guard action	To	Test
1	Idle	eligible search	Searching	TC-SM05-01
2	Searching	server assigns offer	Offered	TC-SM05-02
3	Offered	local accepts before deadline	Accepted	TC-SM05-03
4	Accepted	all humans accept	Ready	TC-SM05-04
5	Offered	local/peer declines	Cancelled	TC-SM05-05
6	Accepted	peer declines	Cancelled	TC-SM05-06
7	Offered	deadline expires	Expired	TC-SM05-07
8	Cancelled	consenting peer policy requeue	Searching	TC-SM05-08
9	Cancelled	decliner leaves queue	Idle	TC-SM05-09
10	Expired	cleanup	Idle	TC-SM05-10
11	Searching	cancel search	Idle	TC-SM05-11
12	Ready	session handed off	Idle	TC-SM05-12
13	Offered	reject foreign/stale acceptance	Offered	TC-SM05-13

SM06 Private room

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F08 / WF04. Evidence: My project/Assets/Scripts/Multiplayer/PhotonSession.cs:116

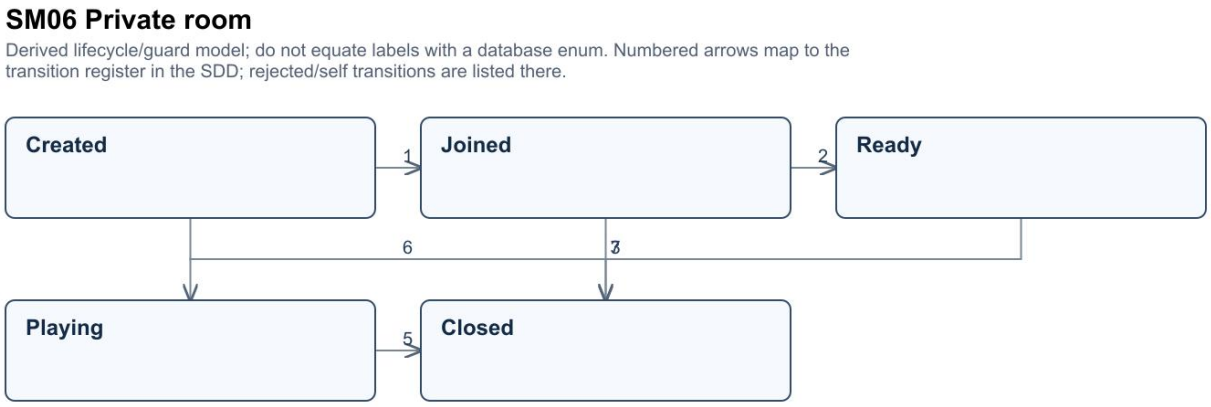


Figure SM06 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Created	eligible guest joins code	Joined	TC-SM06-01
2	Joined	valid teams all ready	Ready	TC-SM06-02
3	Ready	authorized start	Playing	TC-SM06-03
4	Joined	reject non-host/unbalanced start	Joined	TC-SM06-04
5	Playing	session ends	Closed	TC-SM06-05
6	Created	host closes	Closed	TC-SM06-06
7	Joined	room disposed	Closed	TC-SM06-07

SM07 Online canonical match

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F10 / WF03. Evidence: src/Backend/Services/Game/OnlineGameService.OnlineMatches.cs:182

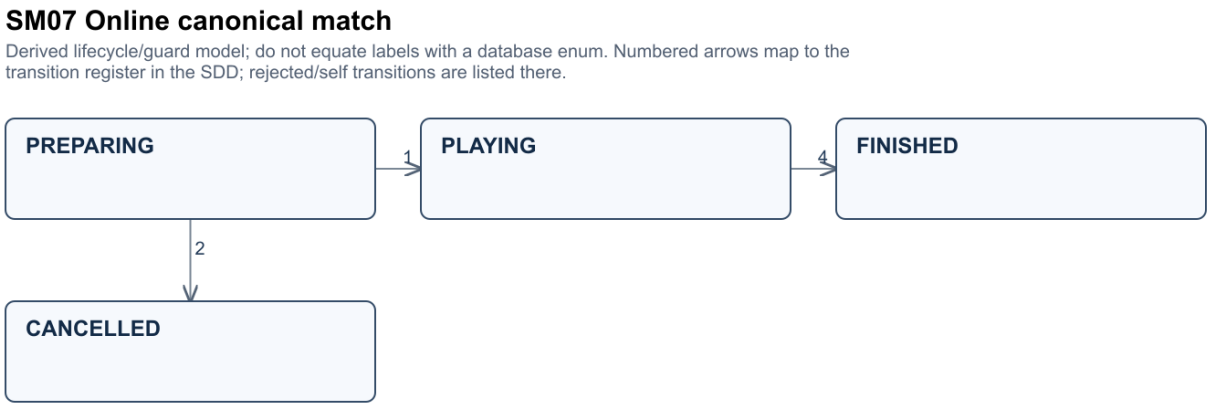


Figure SM07 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	PREPARING	all required humans accepted and start validated	PLAYING	TC-SM07-01
2	PREPARING	decline/timeout	CANCELLED	TC-SM07-02
3	PLAYING	heartbeat or partial report	PLAYING	TC-SM07-03
4	PLAYING	all matching reports or policy forfeit	FINISHED	TC-SM07-04
5	PLAYING	reject conflicting/foreign result	PLAYING	TC-SM07-05
6	FINISHED	duplicate same report returns state	FINISHED	TC-SM07-06
7	CANCELLED	reject late start	CANCELLED	TC-SM07-07

SM08 Reward and command ledger

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F13 / WF05. Evidence: src/Backend/Services/Game/OnlineGameService.cs:90

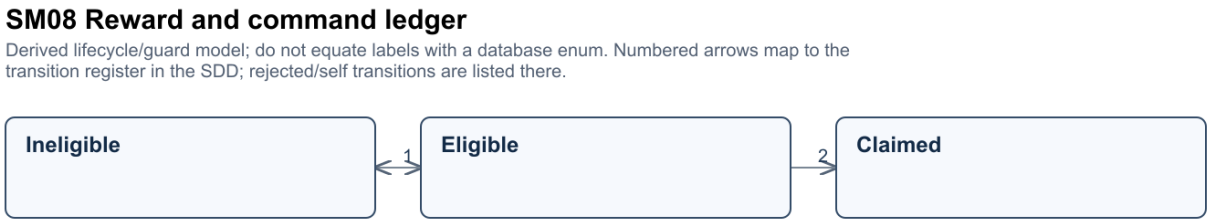


Figure SM08 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Ineligible	objective reaches threshold	Eligible	TC-SM08-01
2	Eligible	valid unique claim commits	Claimed	TC-SM08-02
3	Claimed	same request payload idempotent	Claimed	TC-SM08-03
4	Claimed	reject conflicting replay	Claimed	TC-SM08-04
5	Claimed	new applicable period only	Ineligible	TC-SM08-05

SM09 Friendship

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F12 / WF04. Evidence: src/Backend/Services/Game/OnlineSocialService.cs:27

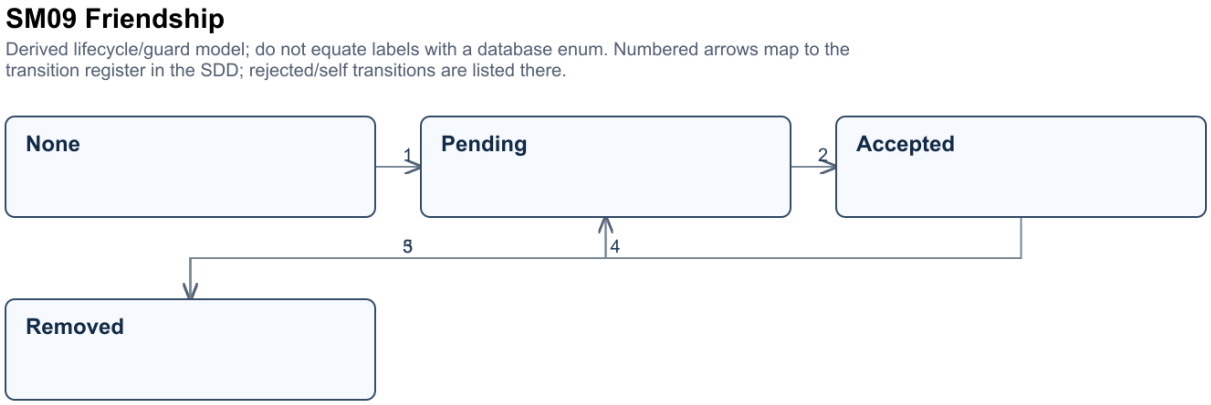


Figure SM09 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	None	valid request	Pending	TC-SM09-01
2	Pending	recipient accepts	Accepted	TC-SM09-02
3	Pending	decline/cancel	Removed	TC-SM09-03
4	Accepted	participant removes	Removed	TC-SM09-04
5	Removed	new valid request	Pending	TC-SM09-05
6	Pending	reject third-party acceptance	Pending	TC-SM09-06

SM10 Replay upload

Derived lifecycle/guard model; do not equate labels with a database enum. Trace F11 / WF05. Evidence: src/Backend/Services/Game/OnlineReplayService.cs:14

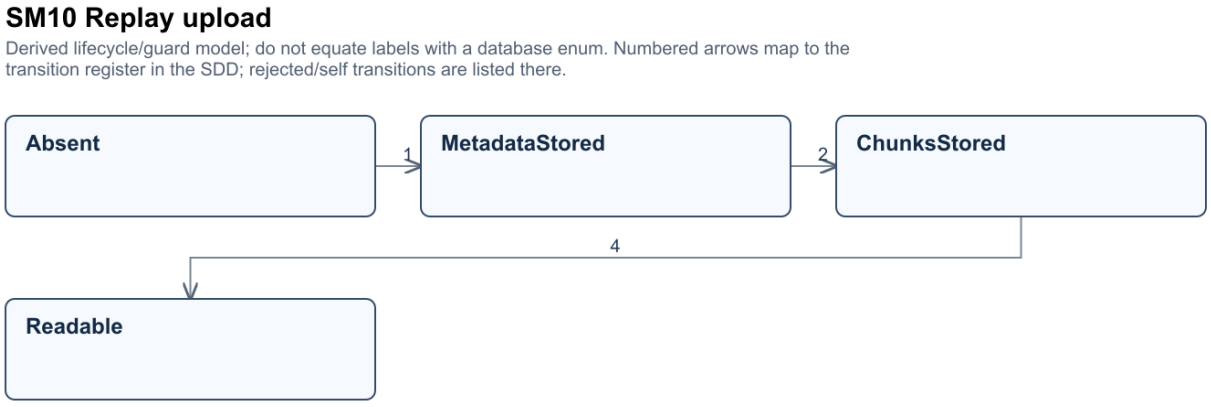


Figure SM10 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Absent	authorized bounded metadata	MetadataStored	TC-SM10-01
2	MetadataStored	valid chunk upload	ChunksStored	TC-SM10-02
3	ChunksStored	same chunk hash idempotent	ChunksStored	TC-SM10-03
4	ChunksStored	required data available	Readable	TC-SM10-04

#	From	Trigger and guard action	To	Test
5	MetadataStored	reject foreign or oversized chunk	MetadataStored	TC-SM10-05

SM11 Administrative sanction

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Trace F15 / WF06. Evidence: src/Backend/Services/Implementations/AdminService.cs:82

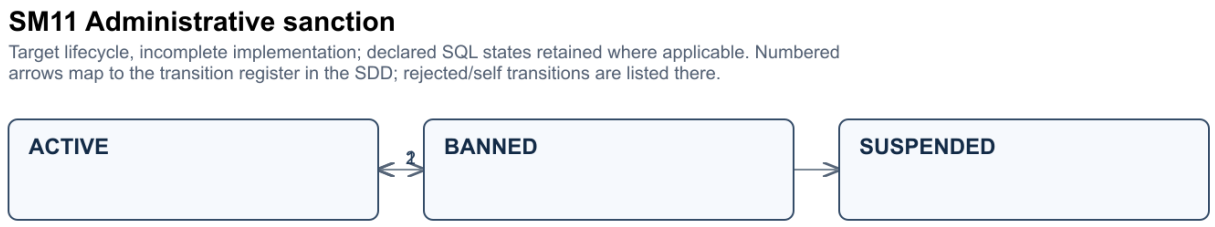


Figure SM11 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	ACTIVE	authorized ban with SQL-valid value	BANNED	TC-SM11-01
2	BANNED	authorized restore with SQL-valid value	ACTIVE	TC-SM11-02
3	ACTIVE	authorized suspension target	SUSPENDED	TC-SM11-03
4	SUSPENDED	authorized restore target	ACTIVE	TC-SM11-04
5	ACTIVE	reject non-admin mutation	ACTIVE	TC-SM11-05

SM12 Lesson publication target

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Trace F16 / WF06. Evidence: src/Backend/Services/Implementations/AdminService.cs:159

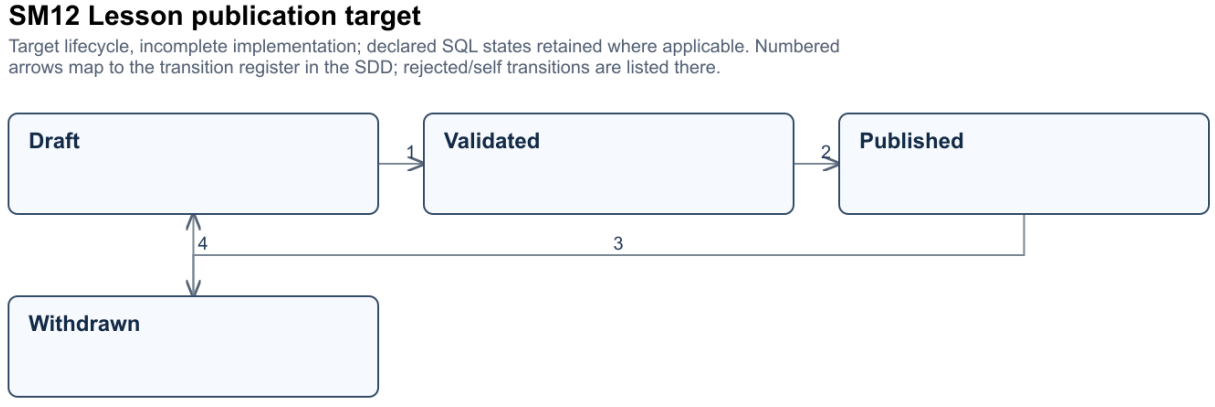


Figure SM12 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Draft	validate code/config and lesson/drill links	Validated	TC-SM12-01
2	Validated	authorized publication updates catalog version	Published	TC-SM12-02
3	Published	unpublish target	Withdrawn	TC-SM12-03
4	Withdrawn	edit target	Draft	TC-SM12-04
5	Draft	reject incomplete content	Draft	TC-SM12-05

SM13 Moderation and appeal target

Target lifecycle, incomplete implementation; declared SQL states retained where applicable. Trace F17 / WF06. Evidence: src/Backend/API/Controllers/AdminController.cs:10

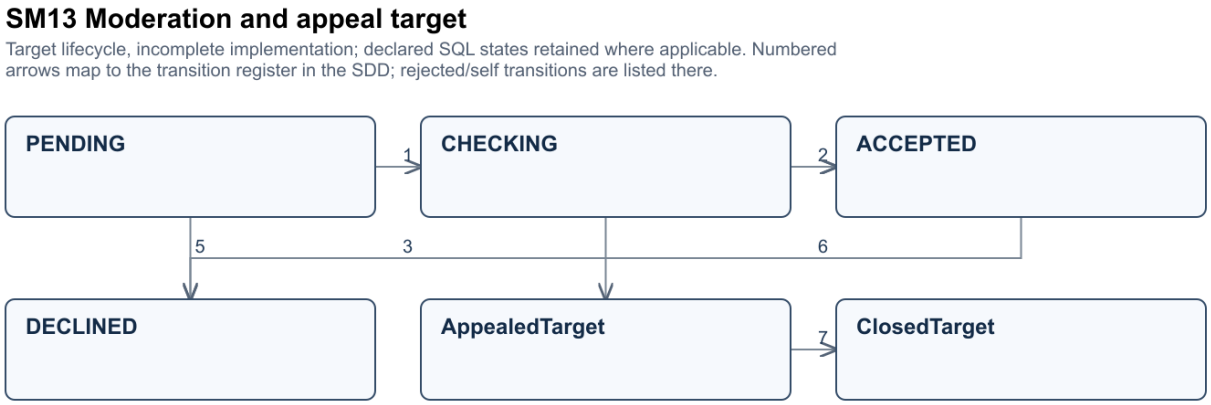


Figure SM13 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	PENDING	authorized review begins	CHECKING	TC-SM13-01
2	CHECKING	valid accepted decision	ACCEPTED	TC-SM13-02
3	CHECKING	valid declined decision	DECLINED	TC-SM13-03
4	PENDING	direct valid decision allowed by current setter	ACCEPTED	TC-SM13-04
5	PENDING	direct valid decision allowed by current setter	DECLINED	TC-SM13-05
6	ACCEPTED	eligible appeal target	AppealedTarget	TC-SM13-06
7	AppealedTarget	appeal resolved target	ClosedTarget	TC-SM13-07
8	PENDING	reject unauthorized decision	PENDING	TC-SM13-08

SM14 Training controller phase

Actual controller enum; transitions are selected main control paths. Trace F03 / WF01. Evidence: My project/Assets/Scripts/LocalMatch/TrainingDirector.cs:14

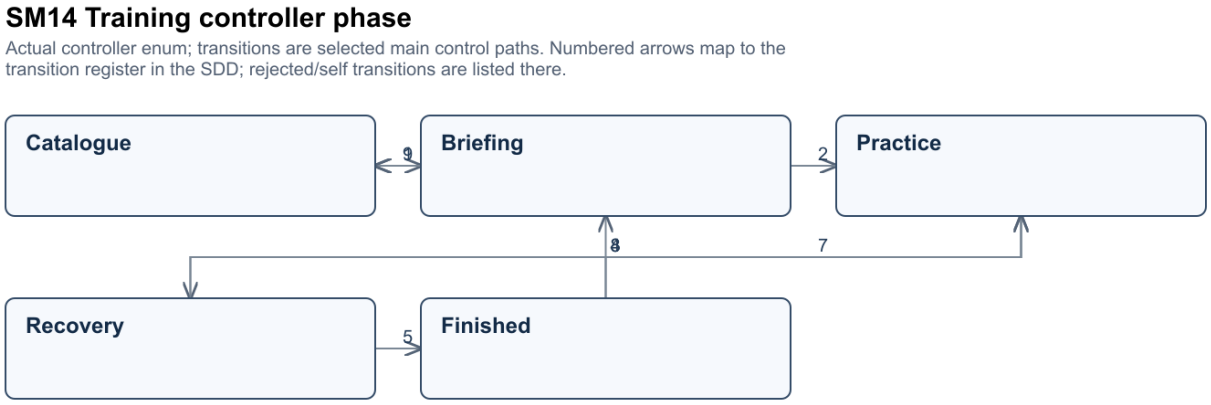


Figure SM14 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Catalogue	select unlocked lesson	Briefing	TC-SM14-01
2	Briefing	server attempt start succeeds	Practice	TC-SM14-02

#	From	Trigger and guard action	To	Test
3	Practice	trial success or error	Recovery	TC-SM14-03
4	Recovery	prepare next trial	Practice	TC-SM14-04
5	Recovery	all trials complete and submit result	Finished	TC-SM14-05
6	Practice	return to catalog and save	Catalogue	TC-SM14-06
7	Finished	retry lesson	Practice	TC-SM14-07
8	Finished	select next lesson	Briefing	TC-SM14-08
9	Briefing	return to catalog	Catalogue	TC-SM14-09
10	Briefing	start request fails; remain and show error	Briefing	TC-SM14-10

SM15 Multiplayer client flow

Actual FlowState enum; ready acceptance is a separate canonical match state. Trace F09 / WF03. Evidence: My project/Assets/Scripts/Entites/Multiplayer/RoomModels.cs:7

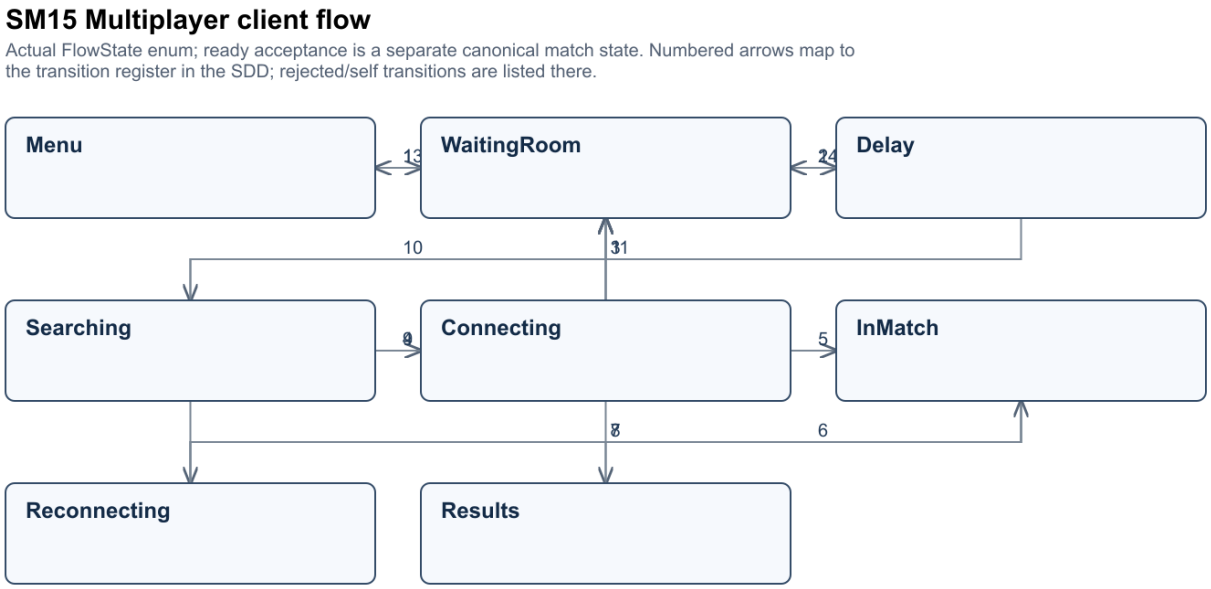


Figure SM15 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

#	From	Trigger and guard action	To	Test
1	Menu	open/join valid room	WaitingRoom	TC-SM15-01
2	WaitingRoom	begin eligible search	Delay	TC-SM15-02
3	Delay	queue begins	Searching	TC-SM15-03
4	Searching	assigned match ready offer	Connecting	TC-SM15-04
5	Connecting	court and required peers ready	InMatch	TC-SM15-05
6	InMatch	match result shown	Results	TC-SM15-06
7	InMatch	connection interrupted	Reconnecting	TC-SM15-07
8	Reconnecting	rejoin succeeds	InMatch	TC-SM15-08
9	Reconnecting	grace fails	WaitingRoom	TC-SM15-09
10	Searching	cancel search	WaitingRoom	TC-SM15-10
11	Connecting	ready offer cancelled	WaitingRoom	TC-SM15-11
12	Results	return to room	WaitingRoom	TC-SM15-12
13	WaitingRoom	return or sign out	Menu	TC-SM15-13
14	Delay	search setup fails	WaitingRoom	TC-SM15-14

6 Main workflow activity design

WF01 Register learn and complete a first drill

Actors: Player, SMTP delivery service. Proposed owner: M1. Features: F01, F02, F03.

Account registered → verified → tutorial completed → training attempt completed.

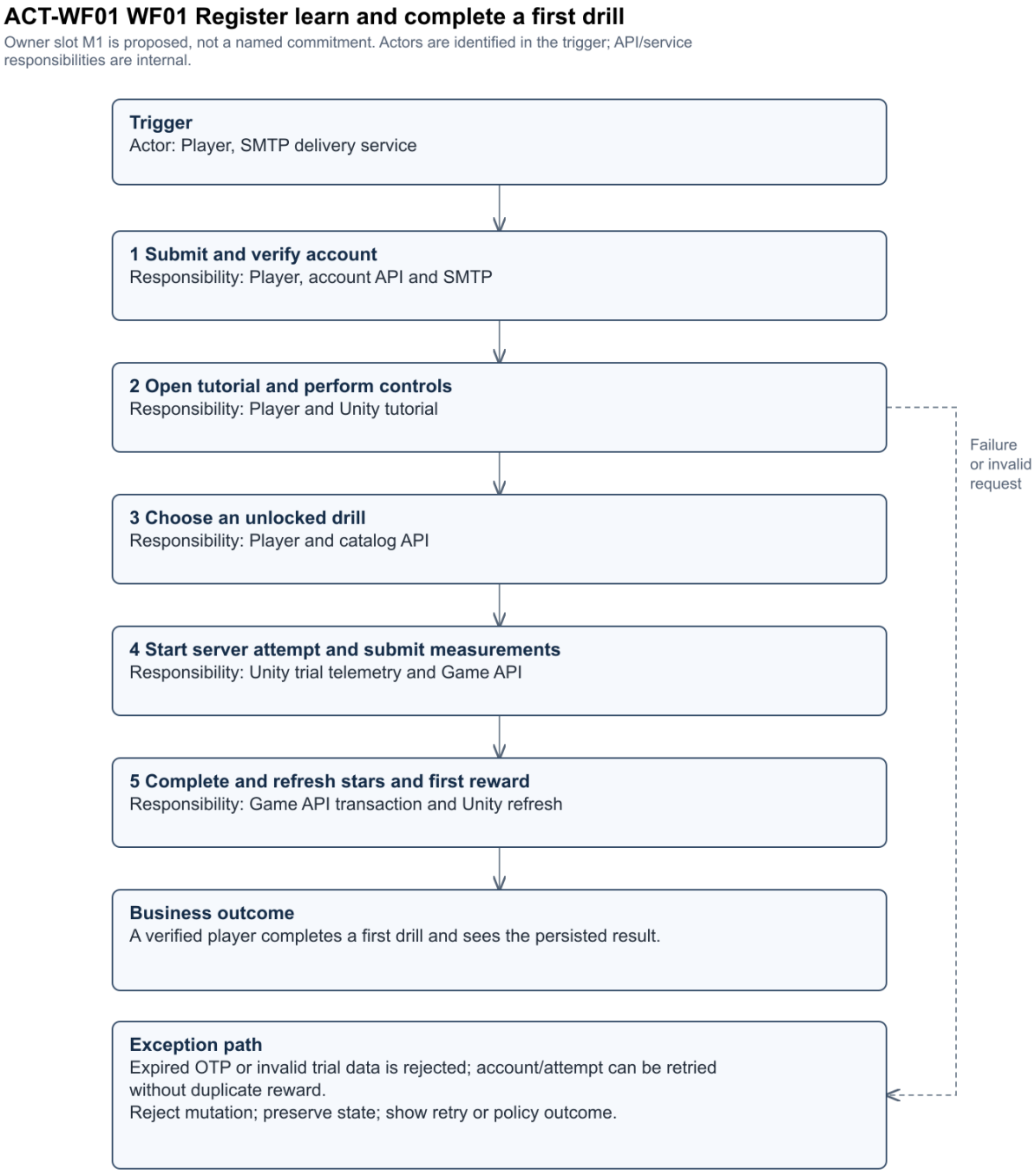


Figure ACT-WF01 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception Expired OTP or invalid trial data is rejected; account/attempt can be retried without duplicate reward.

Demo WF01-MAIN and WF01-EXCEPTION. End-to-end case TC-WF01. Required outcome: A verified player completes a first drill and sees the persisted result.

WF02 Practice against an adaptive bot

Actors: Player. Proposed owner: M2. Features: F05, F06, F10, F11.

Match Intro → PreServe → Rally → MatchOver → result persisted.

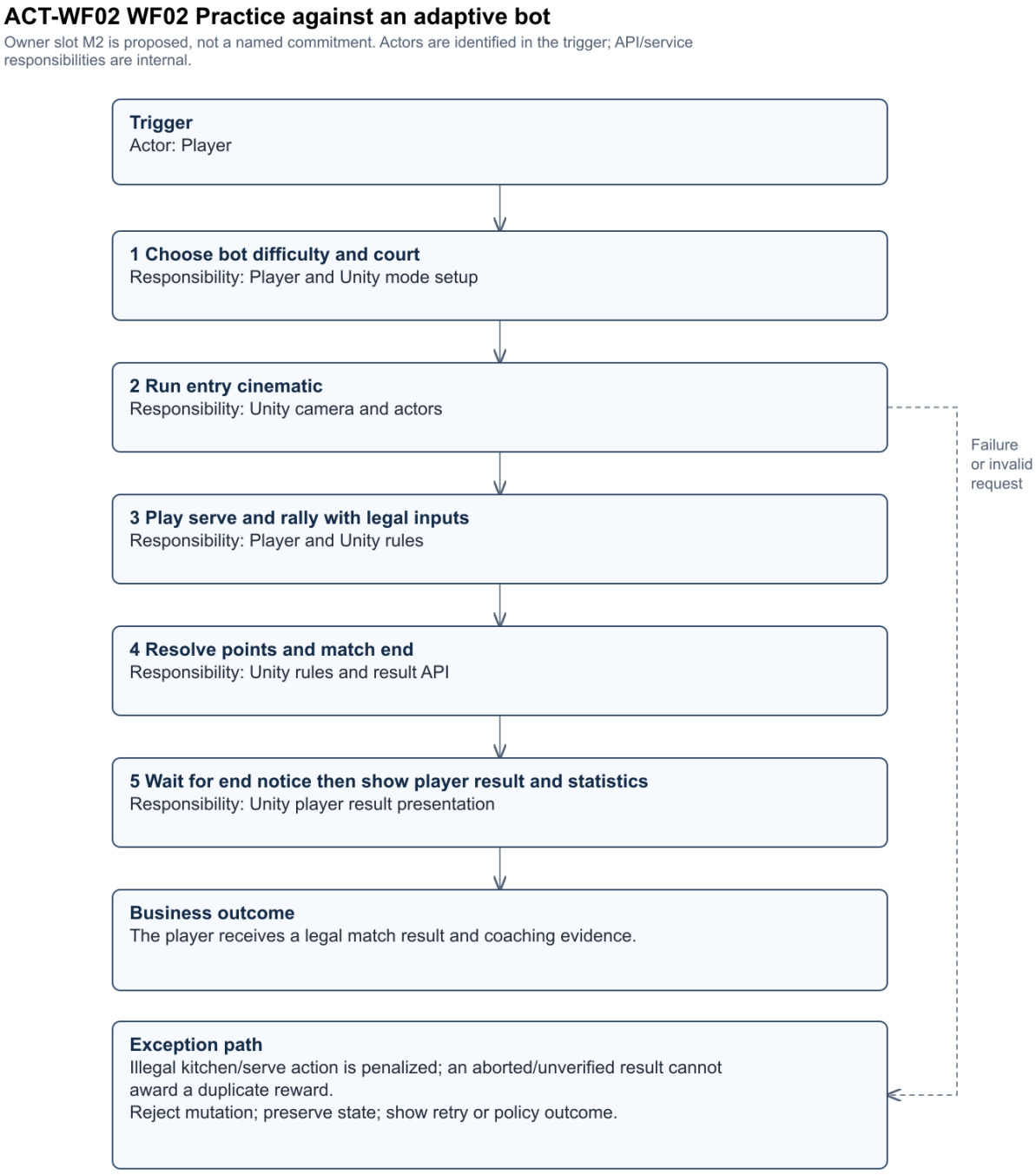


Figure ACT-WF02 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception Illegal kitchen/serve action is penalized; an aborted/unverified result cannot award a duplicate reward.

Demo WF02-MAIN and WF02-EXCEPTION. End-to-end case TC-WF02. Required outcome: The player receives a legal match result and coaching evidence.

WF03 Find accept and finish an online match

Actors: Online player A, Online player B, Photon service. Proposed owner: M3. Features: F06, F07, F09, F10.

Queue searching → offer → all accepted → playing → reported → finished.

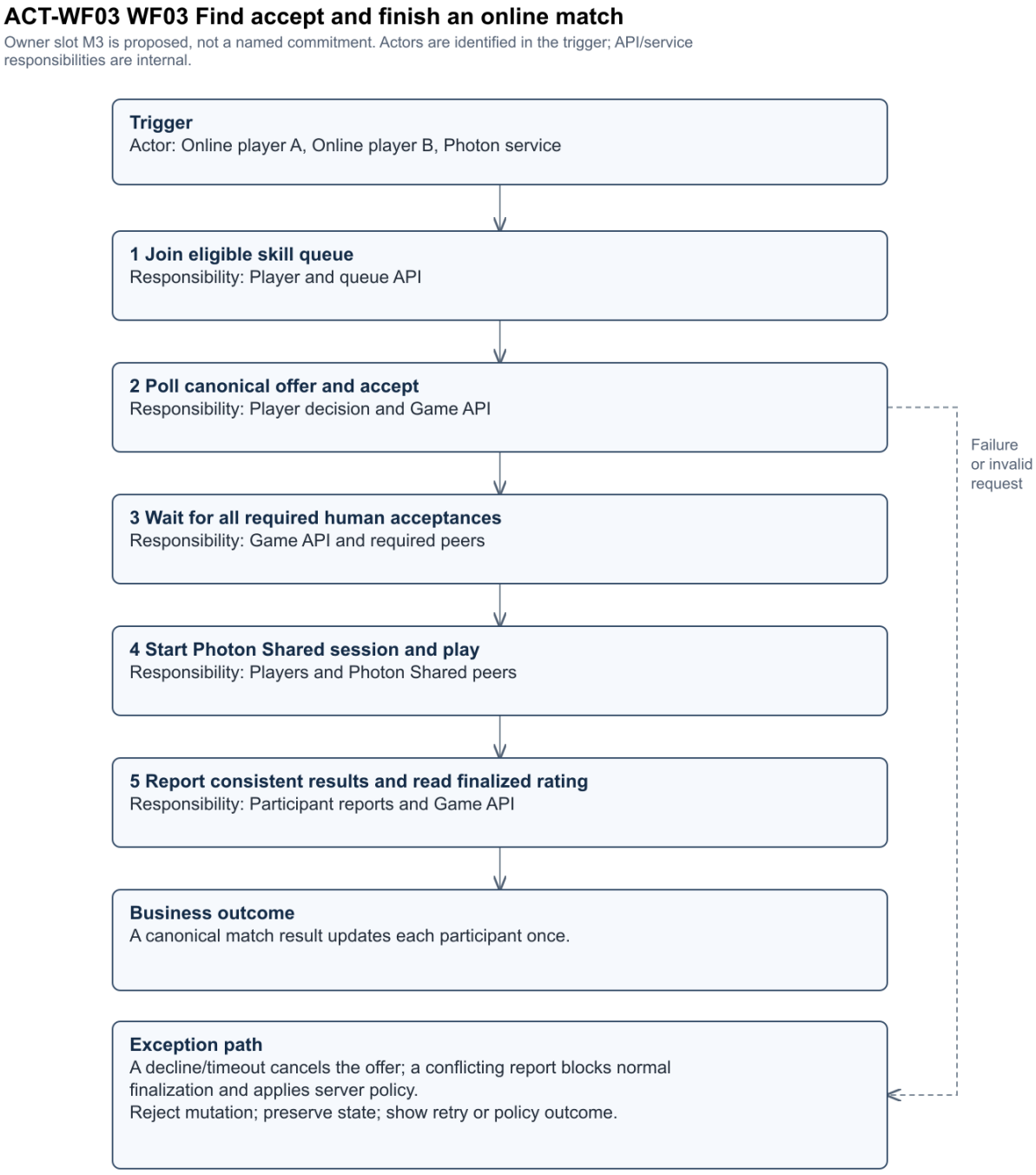


Figure ACT-WF03 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception A decline/timeout cancels the offer; a conflicting report blocks normal finalization and applies server policy.

Demo WF03-MAIN and WF03-EXCEPTION. End-to-end case TC-WF03. Required outcome: A canonical match result updates each participant once.

WF04 Create a private team match

Actors: Room host, Invited player. Proposed owner: M3. Features: F08, F09, F10, F12.

Room created → joined → teams ready → playing → finished.

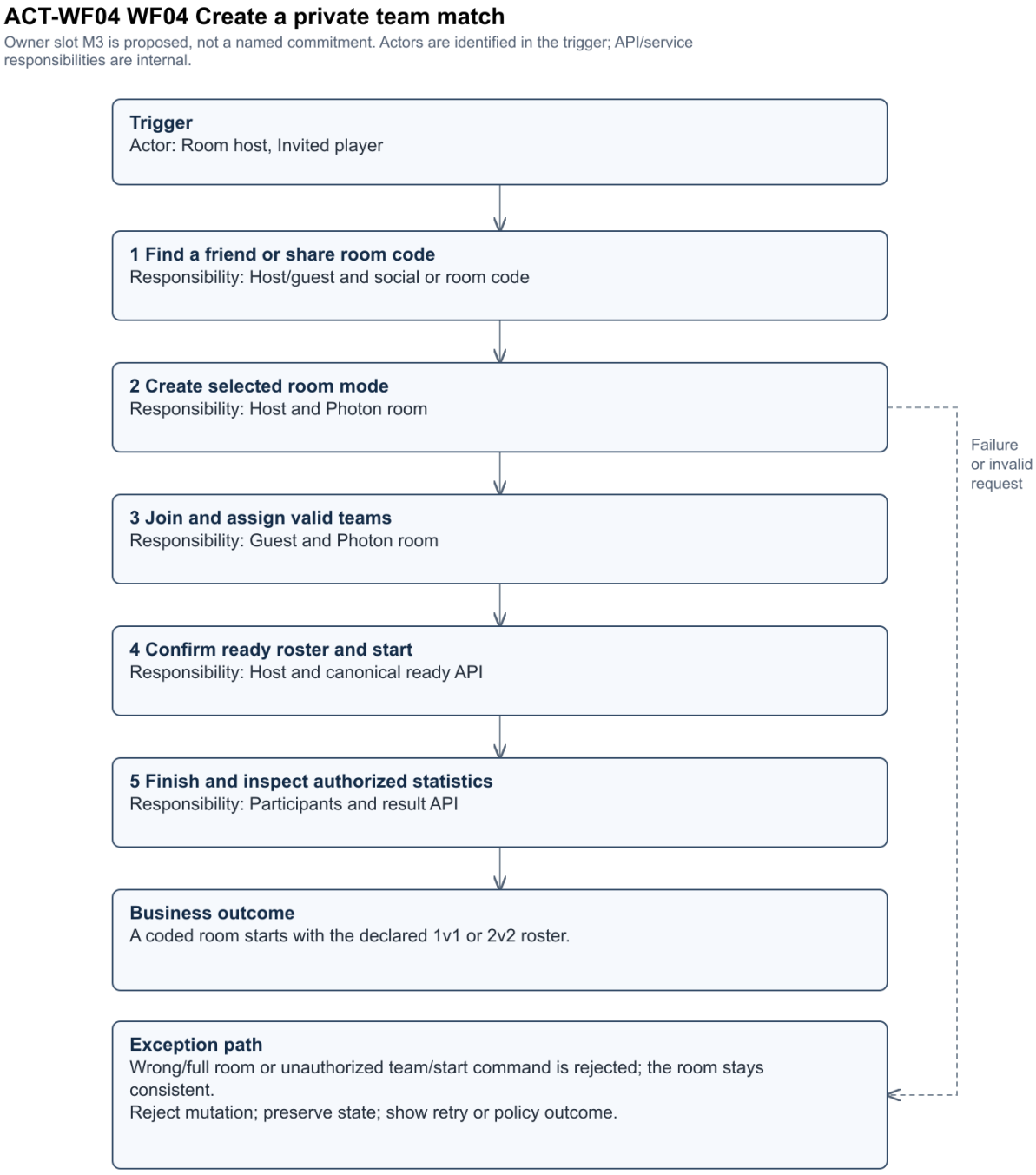


Figure ACT-WF04 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception Wrong/full room or unauthorized team/start command is rejected; the room stays consistent.

Demo WF04-MAIN and WF04-EXCEPTION. End-to-end case TC-WF04. Required outcome: A coded room starts with the declared 1v1 or 2v2 roster.

WF05 Review progress and claim rewards

Actors: Player. Proposed owner: M1. Features: F03, F04, F11, F13, F14.

Attempt completed → progress refreshed → reward eligible → claimed.

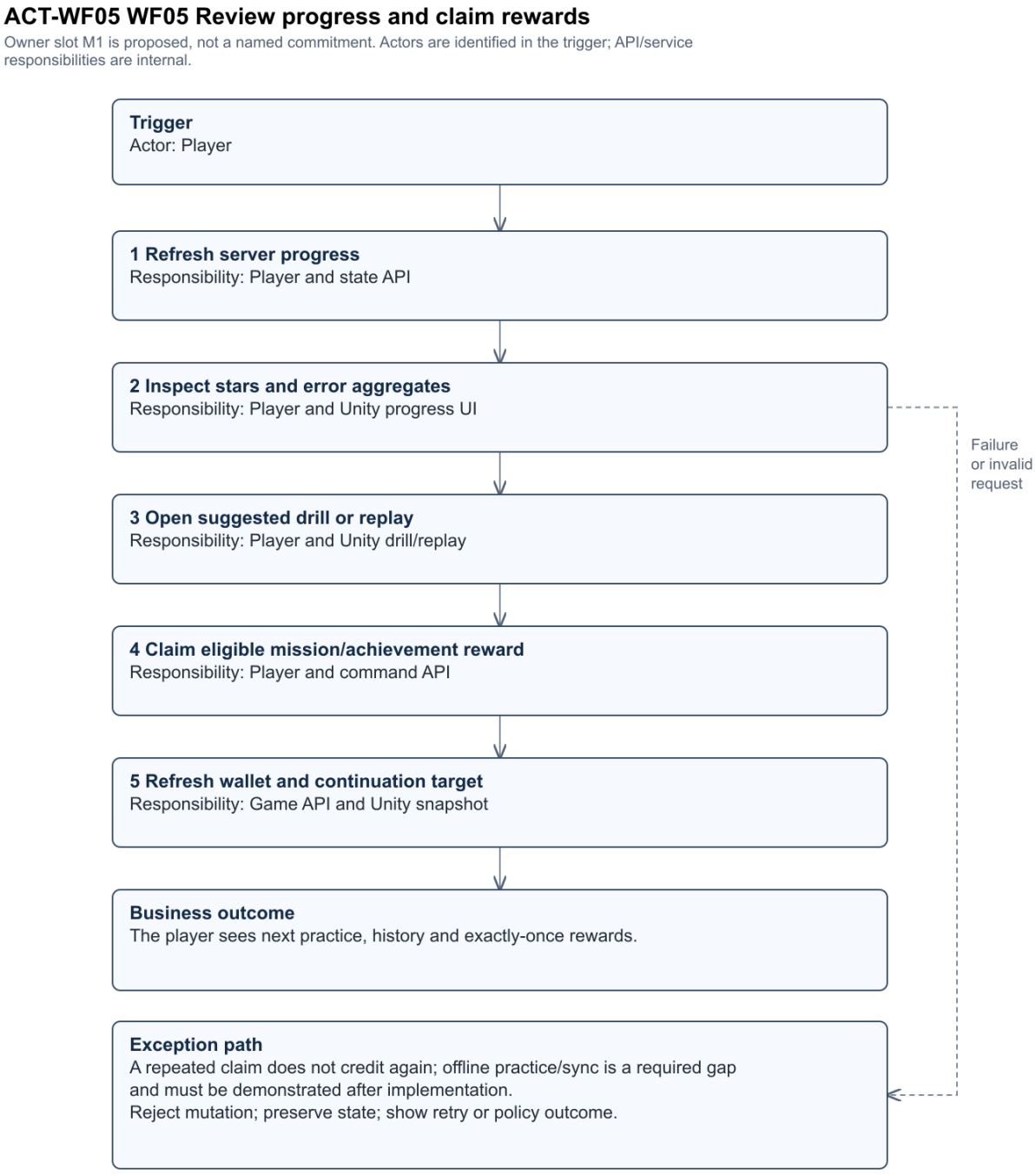


Figure ACT-WF05 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception A repeated claim does not credit again; offline practice/sync is a required gap and must be demonstrated after implementation.

Demo WF05-MAIN and WF05-EXCEPTION. End-to-end case TC-WF05. Required outcome: The player sees next practice, history and exactly-once rewards.

WF06 Administer users content and operations

Actors: Administrator, Reporting player. Proposed owner: M4. Features: F15, F16, F17, F18.

User active → sanctioned/restored; lesson draft → published; report pending → resolved.

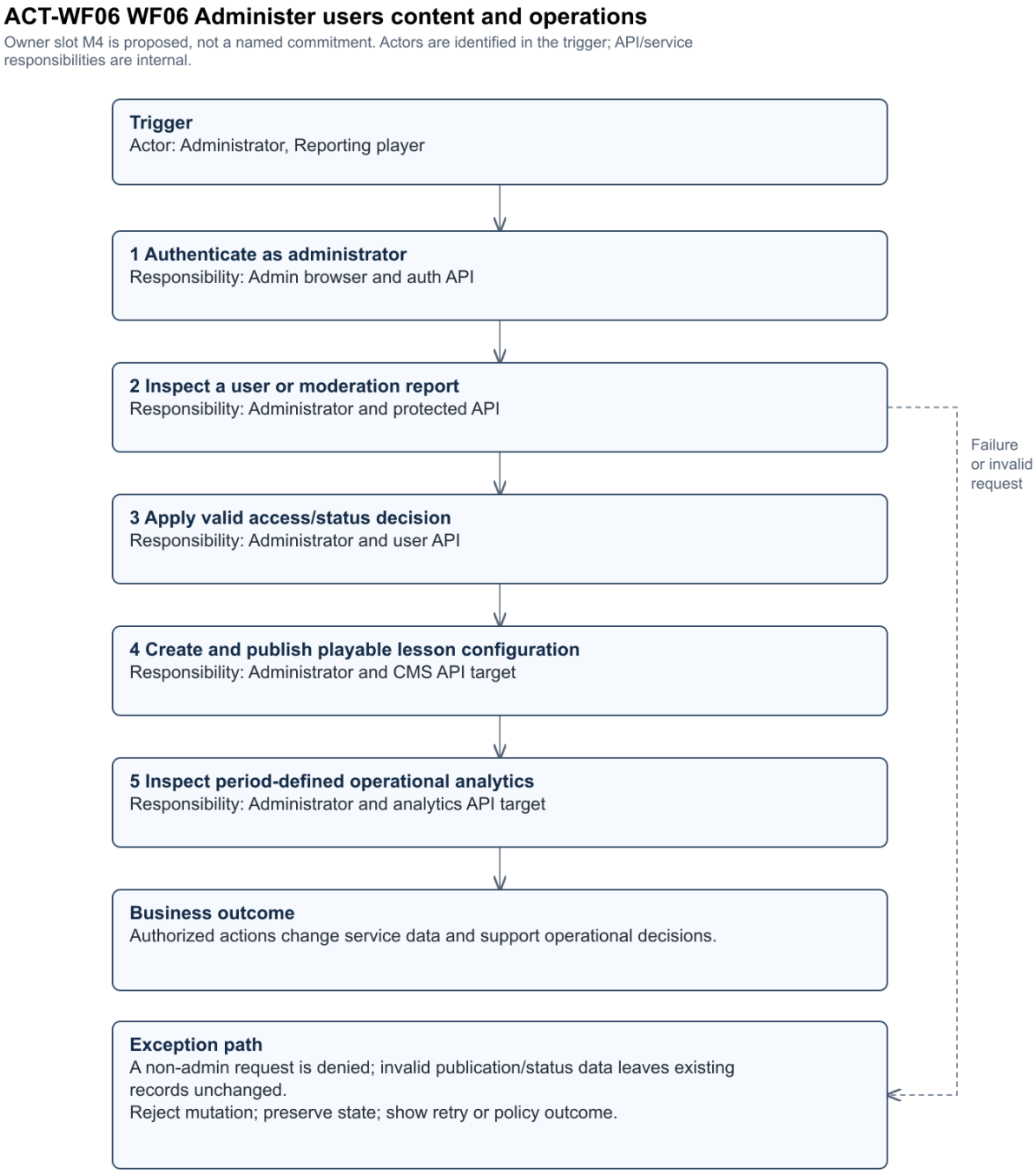


Figure ACT-WF06 · Editable source and full size SVG/PNG are in Diagrams/Current_c32f2a4.

Exception A non-admin request is denied; invalid publication/status data leaves existing records unchanged.

Demo WF06-MAIN and WF06-EXCEPTION. End-to-end case TC-WF06. Required outcome: Authorized actions change service data and support operational decisions.

7 API and shared data contracts

API_Endpoint_Inventory.csv is the complete 81-declaration route register. Action/class authorization must be evaluated together; an empty parsed class-level field does not imply an anonymous endpoint. Legacy finish and submit-drill handlers returning 410 are inventory entries, not functioning alternatives to Game commands.

Route family	Current contract
Auth and Profiles	Registration/OTP/login/refresh/logout and own profile. Unity DTO uses Email/Password and reads AccessToken/Profile. Admin browser must be aligned to this contract.
Game catalog/state/claim/purchase/equip/tutorial	JWT account scope; request IDs and server snapshot revisions; catalog rules and authoritative balances.
Game training/start/checkpoint/complete	Server attempt identity, catalog code, monotonic validated measurements and completion award policy.
Game ranked and match/online	Search identity, protocol/region/party, canonical roster, acceptance/deadline, heartbeat, report fingerprint and finalization.
Game social and replays	Paged real accounts/rankings/relationships; participant-authorized metadata/chunk operations.
Admin	Role-scoped users, content, match/moderation and analytics operations; several integration/definition gaps remain.

8 Deployment environments and operations

Source includes a backend Dockerfile and root docker-compose.yml, but no current deployed staging/prod inventory or verified CI workflow is supplied. The following is a proposed operational design, not a claim that it has been deployed.

Environment	Proposed control	Verification required
Development	Local isolated DB/Redis/object-store settings; developer-specific accounts; no committed secrets	Document actual ports, engine versions and API base URLs.
Staging	Versioned API/web/client builds; seeded repeatable Review 2 data; separate Photon/cloud configuration	Run all main/exception workflows and persist run logs tied to commit and schema version.
Production	Approved secrets, TLS, network restrictions, backups, health probes and metrics	Prove Admin RBAC, AES-256 deployment requirement, restore procedure and monthly availability.
Release and rollback	PR review and protected main; immutable version tags; reversible schema plan and previous images	Back up before migration, smoke-test candidate, monitor and restore prior image/config on failure; test DB compatibility separately.

Proposed CI: static checks and builds → pure rule/unit tests → isolated PostgreSQL/Redis API integration tests → generated coverage/contract evidence → staging smoke workflows. Do not claim ≥50% automated-in-CI until at least the agreed unique case count has actually run and logs are linked. The IntegrationChecks console source is a candidate harness, not a recorded passing run.

9 Security errors and observability

Use memory tokens on Unity, hashed passwords and rotating refresh tokens, constrained OTP attempts, server-derived owner IDs, transactional idempotency and bounded replay decoding. Verify TLS, secret injection and production defaults. No key, password or connection-string value is reproduced in this pack. Bind Photon identity to authenticated backend identity if stronger peer authentication is required; AuthenticationValues(playerId) alone is not proof of a custom JWT verification service.

Define correlation IDs for commands/matches, redact credentials and personal data in logs, measure queue wait/acceptance/abandonment, result conflicts, training rejection reasons, API latency and DB/Redis health. Derive active users from dated activity sessions, not a total-user count. Define retention cohorts and denominators before presenting charts.

10 AI tooling and documentation limits

Codex, GPT-6 family, assisted read-only repository analysis, requirement tracing, diagram generation and document preparation. The exact UI model identifier was not exposed and requires team confirmation. No source code was authored or changed in this task. Human review must validate sporting semantics, member ownership, scope decisions and test evidence. In-game ML models and their quality are separate from AI assistance used to write documentation.

11 Open design findings

SR01 High Photon Fusion Shared gives a peer state authority. This is not the proposed trusted dedicated simulation and leaves host/peer trust risk. Evidence: My project/Assets/Scripts/Multiplayer/PhotonSession.cs:116

SR02 High The active progression runtime requires server state. Offline drill execution and exactly-once later synchronization are not demonstrated. Evidence: My project/Assets/Scripts/Progression/ProgressionRuntime.cs:8

SR03 High Initial-placement ML, telemetry technique classifier, labelled evaluation and validated recommendations are not evidenced. Rule-based feedback does not establish those commitments. Evidence: My project/Assets/Scripts/LocalMatch/BotMatch.cs:11

SR04 High The React login sends emailOrUsername/passwordHash and reads role/token, while the API requires Email/Password and returns AccessToken/Profile. The default contracts are incompatible. Evidence: src/Frontend/src/context/AuthContext.tsx:41

SR05 High Admin mutations serialize enum status as Banned/Active while the SQL constraint uses ACTIVE/BANNED/SUSPENDED; these values can fail the database constraint. Evidence: src/Backend/Services/Implementations/AdminService.cs:82

SR06 High Admin lesson creation omits catalog Code/Config, while the game catalog selects coded configured lessons. CRUD does not yet demonstrate publishing a playable drill. Evidence: src/Backend/Services/Implementations/AdminService.cs:159

SR07 High Complete live-match oversight, appeals and operational analytics are not evidenced. ActiveUsersToday currently counts all users rather than dated activity. Evidence: src/Backend/Services/Implementations/AdminService.cs:353

SR08 High Current-revision physical-device, network/load, security, availability, offline and ML acceptance results are unavailable. Test source is not execution evidence. Evidence: src/Database.sql

SR09 High The local review-2 tag and CI execution evidence are absent; the required official tagged self-review and automated-in-CI target cannot be verified. Evidence: src/Database.sql

SR10 Medium The approved registration revision, official project code and decisions accepting stack/authority deviations are unconfirmed. No scope reduction has been assumed. Evidence: src/Database.sql

SR11 Medium Server-owned transactions and matching human reports improve result integrity, but peers still supply simulation outcomes and training measurements. Agreement does not independently prove legal physics. Evidence: src/Backend/Services/Game/OnlineGameService.OnlineMatches.cs:182

SR12 Medium Real friendship APIs and room-code play exist and cover the proposal room-code requirement. Account-targeted invitations are an unverified desired UX extension; confirm whether they should become a separate commitment. Evidence: src/Backend/Services/Game/OnlineSocialService.cs:27

SR13 Medium Fallback authentication configuration, deployed TLS/AES-256 controls, production rollback and iOS build acceptance require deployment evidence; configuration values are intentionally redacted. Evidence: My project/Assets/Scripts/Services/Api/UnityApiTransport.cs:17